Paediatric Urology 1

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 41	Management Strategies for Vesicoureteral Reflux
Ch 49	Anomalies of the Kidney and Collecting System
Ch 50	Renal Dysgenesis and Cystic Disease of the Kidney
Ch 51	Pathophysiology of Urinary Tract Obstruction
Ch 52	Ectopic Ureter, Ureterocele and Ureteral Anomalies
Ch 53	Surgery of the Ureter in Children
Ch 56	Hypospadias
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Ch 61	Urologic Considerations in Pediatric Renal Transplantation
Ch 63	Paediatric Urologic Oncology: Renal and Adrenal
Ch 64	Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia
★	Putting it together: common paediatric decisions

41 Management Strategies for Vesicoureteral Reflux

1 · Definition and concept

- **VUR**: retrograde urine flow to upper tract; linked to **acquired scarring** and **congenital dysplasia**; a spectrum needing risk-stratified management.
- **Infection, not reflux, is the injury**: VUR carries infected urine to the kidney. Targets: sterile urine, good emptying.

2 · Embryology and antireflux mechanism

- P1 P4 Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management — *Mackie–Stephens bud position (early bud → high lateral orifice, VUR + dysplasia; late bud → ectopia), passive flap valve, Paquin **5:1** vs 1.4:1*
- Opening is active (longitudinal tunnel muscle shortens/widens the tunnel); closure is passive.

3 · Primary vs secondary VUR

- P4 VUR: etiopathogenesis, recent trends in management and STING technique — *primary (deficient UVJ) vs secondary (PUV, neurogenic, dysfunctional voiding, ureterocele, duplication, diverticulum); overlap; detrusor overactivity commonest urodynamic finding; infant male high voiding pressures*
- Spina bifida: VUR in up to **30%** → early CIC, anticholinergic/β3-agonist. Seek occult dysraphism (sacral dimple, hairy patch, cleft anomaly, refractory constipation).
- Normal bladders void at ~100 cm H_2O without reflux. **Reflux at low pressure / early filling = poor resolution.**
- Treating overactivity (timed voiding, biofeedback, anticholinergics, α-blockers) improves resolution.

> **KEY POINT**
> Primary and secondary are not exclusive: "primary" VUR often has a treatable secondary component (overactivity, BBD). Remove it and only true primary VUR remains.

4 · Epidemiology

- VUR in **~30%** of children with UTI; ~17% of those imaged for hydronephrosis. Under 2 y **38.6%** vs 7.6% at 12–21 y.
- Girls ~2× overall; infant males predominate (up to 3:1 at 0–6 months).

5 · Grading — International Reflux Study (1981)

Grade	Ureter	Pelvis and calyces
I	Non-dilated, ureter only	Not reached
II	Not dilated	Reached, no dilatation
III	Mild–moderate dilatation	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate tortuosity	Moderate dilatation
V	Gross dilatation and tortuosity	Gross dilatation; **papillary impressions lost**

- I–II non-dilating; III–V dilating. Interobserver agreement poor (RIVUR readers agreed ~60%).
- Radionuclide cystography (RNC) 3 grades: 1 = I; 2 = II–III; 3 = IV–V.

- **Ureteral diameter ratio (UDR)** = largest ureteric diameter in false pelvis ÷ L1–L3 vertebral distance; predicts resolution and breakthrough UTI (BUTI) better than grade.

6 · Diagnosis

When to image after UTI

- **AAP 2011**, febrile UTI 2–24 months: **US first**; VCUG only if US abnormal or **second febrile UTI**.
- **NICE**: US; VCUG mainly for atypical/recurrent UTI in younger children; none routinely >3 y.
- **Top-down**: DMSA first, VCUG only if abnormal. Normal DMSA → **<1%** chance of high-grade VUR; misses ~1/3 of (mostly low-grade) VUR.

Cystographic methods

- P1 VCUG — *gold standard; gravity ≤70 cm H_2O; early study once urine sterile; cyclic VCUG +12–20%; filling-phase reflux poor prognosis; postvoid pelvic retention → UPJO; RNC and ceVUS alternatives*
- Direct RNC (^{99m}Tc pertechnetate): more sensitive for II–V, **misses grade I**. Indirect RNC: no catheter, toilet-trained only, sensitivity 70–90%. ceVUS: shows intrarenal reflux; grading not standardised.
- Coexisting **UVJ obstruction** (dilated proximal ureter, distal aperistaltic segment, dilute contrast, delayed drainage) → **relative contraindication to endoscopic injection**.
- Routine cystoscopy no role; use just before open surgery. Uroflow + PVR ± EMG useful; invasive urodynamics only if refractory.

Upper tract

- **RBUS**: serial mainstay; cannot exclude VUR.
- **DMSA**: gold standard for cortical defects (sensitivity 98%, specificity 92%); acute photopenia may resolve, persistence = scar.

7 · Congenital dysplasia vs acquired scar

- P4 P1 VUR: etiopathogenesis, recent trends in management and STING technique — *dysplasia (boy, IV–V, antenatal, not preventable) vs acquired scar (girl, febrile UTI, BBD, preventable); polar compound papillae, intrarenal reflux at 2 mmHg; Ransley–Risdon sterile reflux; "big bang" debated*
- Prenatally detected boys: damage confined to IV–V (nearly half of grade V) before any infection.
- Hypertension in **34–38%** of scarred children. **Correcting VUR does not lower BP**; nephrectomy/segmental resection of a small kidney may.

8 · Bladder and bowel dysfunction (BBD)

- **Treat constipation first** (sphincter co-contraction).
- AUA: BBD is a critical modifiable variable → more UTI on CAP and after surgery, less resolution, **lower endoscopic success, but not lower open surgical success**.
- Recurrent UTI off CAP: ~1/2 BBD + VUR, ~1/5 VUR alone, ~1/3 BBD alone.
- **Phimosis**: in boys with VUR, circumcision NNT to prevent one UTI falls from ~111 to **~4**; topical steroid equivalent.

9 · Natural history and resolution

Grade	Spontaneous resolution
I–II	Most; grade II 63–85%
III	~50%; ~90% of resolutions within 4 years
IV–V, bilateral III	Few (infants with IV–V resolve well in year 1)

- Resolution predictors: **low grade, young age, unilateral, no BBD, voiding-only reflux, higher bladder volume at onset, low UDR**; age matters as much as grade (AUA 1997).

10 · Medical management — continuous antibiotic prophylaxis (CAP)

- P3 Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome — *CAP agents by age with doses (TMP/amoxicillin <2 months; TMP-SMX, nitrofurantoin >2 months), Hidas risk groups, indications for correction, 8–10% febrile UTI after stopping CAP*
- **Manage infection risk, not VUR**. Low dose **at night** (longest retention).
- TMP-SMX: sulfonamide displaces bilirubin → neonatal jaundice; rare reversible leukopenia. Nitrofurantoin: least faecal-flora resistance; pulmonary fibrosis, hepatotoxicity rare at low dose.
- **Smellie's rule**: BUTI organism sensitive to prophylactic → non-adherence/low dose; resistant → high residual or dose.
- **Breakthrough febrile UTI on CAP despite good bladder/bowel habits → stop medical management and correct VUR** (confirm urine collection first).

Risk stratification (Hidas)

Factor	Low risk	High risk
Sex / age	Male; >5 y	Female; <1 y
Laterality / grade	Unilateral; I–II	Bilateral; III–V
BBD / scars	No	Yes
Presentation	Prenatal hydronephrosis	Multiple febrile UTIs

- 2-year BUTI on CAP: **8.6% low, 26% intermediate, 62.5% high**. ~2/3 of children are low risk.

11 · Practical algorithm (AUA-based, risk-stratified)

ALGORITHM

True febrile UTIs + RBUS → VCUG if multiple febrile UTIs, admissions or upper-tract findings (afebrile/doubtful UTI, normal US → conservative)

All: treat BBD (constipation first), hygiene, timed voiding

Low risk (grade I–II, no BBD, male, older) → observation ± CAP; treat febrile UTI promptly

High risk (grade ≥III, girl, BBD, febrile presentation, scars, infant) → CAP

Breakthrough febrile UTI, new scars, high grade with poor function, persistent VUR with low infection tolerance → correction (endoscopic or reimplant)

- Little CAP benefit at grade ≤II without BBD; benefit at ≥III, at least in girls. Scarring, poor function or dysplasia → favour correction.

- Unresolved: asymptomatic bilateral high-grade VUR in older boys; correction in girls before pregnancy.

12 · Surgical correction

- Open correction **>95%**; robotic probably less morbid. Not superior to medical therapy for scarring or growth (IRSC, Birmingham).
- After correction almost no new scars, but adult UTI risk persists.
- **Endoscopic Dx/HA**: BBD lowers success; avoid with UVJ obstruction or obstructive refluxing megaureter. Mounds calcify in over half and **mimic ureteric stones**.
- Techniques and complications *→ Ch 53*. After cross-trigonal reimplant, **retrograde catheterisation and ureteroscopy are difficult**.

13 · Associated conditions

- **UPJO + VUR**: dilated ureter, little/dilute pelvic contrast, no postvoid pelvic drainage. **Correct obstruction first**; simultaneous open repair risks ureteric blood supply.
- **Duplex**: VUR usually into **lower pole** (Weigert–Meyer); duplication alone does not raise risk.
- **Paraureteral diverticulum** resolves like primary VUR; **ureter entering diverticulum** (no backing) will not resolve.
- MCDK: contralateral VUR ~1/4, usually resolves.
- **Megacystis–megaureter**: massive VUR recycles urine → large thin bladder, normal urethra; reimplant, not outlet surgery.
- **Obstructive refluxing megaureter**: ureteric contrast more dilute than bladder; avoid injection; reimplant ± tapering.

14 · Sibling screening

- Sibling prevalence ~**30%** (up to 50%), higher in younger siblings.
- Target is **renal damage, not VUR**. AUA: **VCUG for siblings with cortical abnormality or renal asymmetry on US, or UTI history**; otherwise observe.

15 · Follow-up and adulthood

- Lifelong BP, proteinuria, renal function if scarred.
- **Pregnancy**: VUR → pyelonephritis ~1/3; scars → ~3× hypertension, **~8× pre-eclampsia**. **Scarring, not VUR, drives morbidity.** Reimplanted women still at UTI risk; no excess miscarriage.

16 · Landmark trials

Trial	Design	Result	Take-home
IRSC	<9 y, high grade; CAP vs open surgery	UTI equal (38%); surgery fewer pyelonephritis; scars, 10-y function equal	Surgery ≈ medical for scarring
Birmingham Reflux Study	High grade; medical vs surgical, 5 y	New scars equal, all within 2 y	Equal; supports "big bang"

Trial	Design	Result	Take-home
Swedish Reflux Trial	1–2 y, grade III–IV; CAP vs endoscopic vs surveillance	Girls febrile UTI: 19% CAP, 23% endoscopic, **57%** surveillance; no new scars on CAP; boys no difference	Treat girls with dilating VUR
RIVUR	607 children, grade I–IV after first/second UTI; TMP-SMX vs placebo	Recurrent UTI **12.9% vs 23.6%**; scarring unchanged; most benefit with febrile index UTI/BBD	CAP prevents UTI, not scars
PRIVENT	576 children with UTI; CAP vs placebo	Absolute UTI reduction 6% (19% → 13%)	Modest benefit
French / low-risk trials	Grade I–III; CAP vs none	No significant difference	No benefit at low risk

KEY POINT

Trials disagree because of population risk: RIVUR was diluted by low-risk children (~8% grade IV); reanalysis NNT 5 high-risk vs 18 low-risk. CAP cuts infection in proportion to baseline risk; neither CAP nor surgery prevents scarring better.

49 Anomalies of the Kidney and Collecting System

1 · Embryology that explains everything

- Ureteric bud (UB) arises from the caudal wolffian duct (WD) at **5–7 wk**, invades the metanephric mesenchyme (MM) and branches; UB tips induce nephrons (**reciprocal induction**).
- Ascent with 90° ventromedial rotation, weeks 6–9: calyces lateral, pelvis medial.
- Renal and genital development are integrated → renal anomalies travel with WD (male) and müllerian duct (MD, female) anomalies. Adrenal (cortex mesoderm, medulla neural crest) is independent.

> **KEY POINT**
> Every anomaly of number is a failure of UB–MM induction: timing and extent set the phenotype. An early insult drags the WD/MD with it (genital anomalies). Arrested ascent leaves an unrotated kidney with anterior pelvis and fetal blood supply; fusion freezes kidneys where they meet.

2 · Bilateral renal agenesis (BRA)

- **≈1 in 5000** births; **75% male**. First-degree relatives ≈1000× risk → **US parents and siblings** of any infant with URA, BRA or dysgenesis.
- **Pathology**: ureters atretic; bladder absent (≈½) or hypoplastic (UB/WD entry into cloaca initiates bladder development). Adrenal flattened — **"lying down" sign**.
- **Potter facies**: senile look, infra-canthal fold, blunted nose, low-set-appearing ears; claw hands, bowed clubbed legs — **deformations** of absent amniotic fluid, not malformations.
- Associated: undescended testes 43%, testicular agenesis ≈10% (vas usually normal); females müllerian/ovarian anomalies; GI anomalies ≈60%.
- **Pulmonary hypoplasia** kills. Fetal urine >90% of amniotic fluid in third trimester. **Peters two-step model**: early kidney-derived "renal factor" (proline) before 16 wk, then amniotic fluid in canalicular phase (16–26 wk) → fluid only partly restores lungs.
- **Diagnosis**: prenatal severe oligohydramnios, no kidneys, small chest. Neonate: **anuria >24 h without bladder distension** = BRA until proved otherwise; US + power Doppler; equivocal → DMSA (no uptake anywhere).
- **Prognosis**: ≈40% stillborn; rest die in 24–48 h of respiratory failure. Amnioinfusion experimental (RAFT, §9): only under IRB oversight with full counselling incl. palliative care.

> **KEY POINT**
> BRA is lethal through the lungs, not the kidneys. Amnioinfusion replaces only the amniotic step, so it converts pulmonary deaths into neonatal ESRD.

3 · Unilateral renal agenesis (URA) and the solitary functioning kidney (SFK)

- **1 in 1200**; male 1.8:1; left slightly more. Many apparent URAs are **involuted MCDK** (curvilinear calcification on CT; colonic flexure in normal place). Renal aplasia may be the commonest cause of congenital solitary kidney. 20-week scan pitfall: adrenal or spleen mistaken for kidney.

- Associations: VACTERL (≈30% have URA), MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia), Turner, Kallmann, branchio-oto-renal, DiGeorge. **Maternal diabetes triples risk**. Isolated preauricular pit/single umbilical artery → no routine US; ≥2 anomalies → US.
- **Ipsilateral ureter absent in ≈60%**; "hemitrigone" is a misnomer (trigone from urogenital sinus). Defect in UB (RET); gonad rarely affected.
- **Contralateral**: VUR **≈30%**, PUJ obstruction ≈11%, VUJ obstruction ≈7%.
- **Genital anomalies 20–40%**: females 25–50%, males 10–15%. Adrenal agenesis <10%.

	Male (WD derivatives)	Female (MD derivatives)
Always present	Testis, head of epididymis	Fimbriated end of tube
Commonest	Absent vas/epididymal body-tail/seminal vesicle (≈50%). **79% of men with absent vas lack the ipsilateral kidney**	Unicornuate/bicornuate, didelphys or septate uterus; vaginal atresia/duplication; Gartner duct cyst with blind ectopic ureter; infertility up to ⅓ with unicornuate uterus
Named syndrome	**Zinner**: seminal vesicle cyst + ipsilateral agenesis	**OHVIRA** (Herlyn–Werner–Wunderlich): obstructed hemivagina, ipsilateral renal anomaly; pubertal haematocolpos, pain
Trigger to image kidneys	Impalpable vas (infertility, herniotomy/orchidopexy)	Any uterine anomaly; pelvic US after thelarche, before menarche

- **Magee classification**: I (before wk 4, whole ridge) → unicornuate uterus; II (early wk 4, WD + UB) → didelphys with obstructed hemivagina on agenesis side; III (after wk 4, UB only) → isolated URA, normal genitalia.
- **MRKH**: 46,XX, upper vaginal to total müllerian agenesis (1 in 4500 females). Type I: renal anomalies uncommon. **Type II (commoner): renal anomalies 40–60%**. MURCS = most severe end.
- **Work-up**: compensatory hypertrophy from 20 wk in ≈90%. US + Doppler; DMSA finds missed ectopic kidney. VCUG for 28% contralateral VUR **not routine** (no fewer febrile UTIs); shared decision.
- **Sport**: not precluded — individual assessment, padding (vehicle accidents cause more renal trauma).

Prognosis and surveillance of the SFK

- **Brenner hyperfiltration**: reduced nephron mass → glomerular hypertension → albuminuria → sclerosis. Highest risk when the SFK itself has CAKUT. Risk surrogates: low birth weight, prematurity, small kidney, diabetic mother.
- KIMONO: ≈⅓ renal injury by ≈10 y; population ≈4× ESRD → lifelong follow-up.

Situation	BP + first-morning urine albumin	Ultrasound	Creatinine
SFK without CAKUT	Annually	Every 5 y if GFR >60 and no medication	Periodic, into adolescence
SFK with CAKUT	Twice yearly	As indicated	Periodic
GFR <60 or on antihypertensive/antiproteinuric	BP, albumin, creatinine 2–4×/year		

Warning signs: hypertension, microalbuminuria (>30 mg/24 h) → ACE inhibitor, salt limitation, avoid excess protein. Hypoplasia/dysplasia *→ Ch 50*.

4 · Supernumerary kidney

- **Own capsule, collecting system and separate blood supply** (vs duplex: one capsule, shared supply). Rare; left; ≈60% caudal to dominant kidney. Ureters converge or separate (≈50% each).
- ≈½ hydronephrotic; obstruction/infection/stone in early adulthood. MRU or retrograde for anatomy, renography for function; cystoscopy shows one or two ipsilateral orifices.

5 · Ectopia (anomalies of ascent)

Simple ectopia

- **1 in 900** (autopsy); bilateral ≈10%. **Pelvic commonest**; lumbar, abdominal, cephalad, thoracic. Aberrant vessels are consequence, not cause.
- Small, lobulated, malrotated (pelvis anterior). **Hydronephrosis common**: ½ obstruction (PUJ ≈70%), ¼ VUR ≥ III, ¼ malrotation. **VUR 30–50%** (often into orthotopic kidney).
- Anomalous blood supply → **map by CT/MRA before surgery**.
- Genital anomalies 15–45%: females 20–66%; males hypospadias, undescended testis ≈5% each. In ≈14% of cloacal malformations. Girl with continuous incontinence → ectopic ureter of the ectopic kidney.
- Empty renal fossa; DMSA/MRU finds a small non-functioning kidney mistaken for URA. Contralateral kidney usually normal; adrenal normally placed.
- **Never remove a "pelvic mass" without imaging** — solitary pelvic kidneys have been excised as tumours.

Cephalad and thoracic kidney

- Cephalad: classic with **omphalocele**.
- Thoracic (≈1 in 13,000): in the **foramen of Bochdalek**, outside pleural space; usually asymptomatic (delayed mesonephric involution; 0.25% of diaphragmatic hernias contain a kidney). Smooth posterior mass on CXR → CT/MRU, avoid thoracotomy; operate only for respiratory distress. Adjacent lower lobe may be hypoplastic.

6 · Fusion anomalies

Crossed renal ectopia

- Kidney opposite its ureteric orifice; **90% fused**; second commonest fusion anomaly. Left-to-right 3:1. Ureter crosses at the pelvic brim; orifice orthotopic.
- **McDonald–McClellan types**: with fusion (90%), without fusion, solitary crossed (imperforate anus ≈20%), bilateral crossed (rarest).

Crossed fused form	Key feature
Inferior ectopia (≈⅔, commonest)	Crossed upper pole fused to normal lower pole; pelves anterior
Sigmoid (S-shaped)	Second commonest; pelves face opposite ways
Lump (cake)	Lobulated pelvic mass
L-shaped (tandem)	Crossed kidney transverse, anterior to L4
Disc (pancake)	Medial fusion, reniform outline kept

Crossed fused form	Key feature
Superior ectopia	Least common; crossed lower pole to normal upper pole

- Pelvis **anterior = early fusion** (before rotation); medial = late. IMA and mesenteric root stop ascent.
- VUR ≈20% (71% bilateral crossed); ectopic orifice ≈3%. Genital: cryptorchidism, hypospadias, absent vas, vaginal atresia.
- Pain, UTI, haematuria, stones in 3rd–4th decade. **Stone in the crossed ureter hurts on the anephric side**; pyelonephritis/PUJ obstruction hurts where the kidneys lie.
- Imaging: US first; diuretic renography for drainage; MRU/MRA before major surgery. Midline-fused kidney with single ureter dividing into two pelves → imperforate anus, vertebral anomalies.

Horseshoe kidney

- **Commonest fusion anomaly**: **1 in 400** (0.25%); male 2:1. **Lower-pole fusion 95%**, before rotation.
- **Isthmus** (parenchyma or fibrous band) at L3/L4, **just below the IMA**, anterior to aorta/IVC.
- Calyces **point posteriorly**, lowest medially (may overlie spine); high ureteric insertion; ureter bends over isthmus. **Blood supply highly variable** → MRA/CT before surgery.
- **Associations** (≈½): **Turner — ≈60% have horseshoe**, caudal regression, neural tube defects (3%), anorectal/VATER; hypospadias/undescended testis ≈4%.
- **PUJ obstruction 13–34%** (high insertion, ureter over isthmus, crossing vessels); many hydronephroses non-obstructed. VUR 8–32%.
- **Stones ≈⅓ of adults**: stasis plus metabolic risk → still do metabolic work-up.
- **Tumours**: **Wilms 1.76–7.9×** (favourable histology; preoperative chemotherapy preserved ≈75% parenchyma — NWTS); pelvic tumours increased; RCC not increased.
- ≥50% asymptomatic; vague abdominal pain to lower lumbar region; UTI ≈30%. US may miss fibrous isthmus; DMSA shows functioning band. Pregnancy unaffected; transplantable.

> **KEY POINT**
> † PUJ obstruction in horseshoe kidney: dismembered pyeloplasty with dependent anterior anastomosis; symphysiotomy not routinely needed.

7 · Malrotation and vascular anomalies

- Malrotation ≈1 in 939; male 2:1; frequent in Turner. **Weyrauch**: **ventral (no rotation, commonest)**; ventromedial; dorsal (180°, rarest); lateral (reverse). **Vessel course is the only clue** to direction.
- Hilar fibrosis → intermittent PUJ obstruction, stones; function unaffected. Flattened elongated pelvis, stretched upper calyx, laterally displaced upper ureter; suspect when a stone lies in an unusual position. Bilateral malrotation mimics horseshoe.
- P2 Anomalies of renal artery. — *multiple / aberrant / accessory terms; single artery 71–85%; end-arteries; crossing vessels in normal PUJs; aneurysm (Abeshouse, treat >2 cm, pregnancy, symptoms); congenital vs acquired AVF, embolisation*
- Aneurysm: ring calcification ≈30%; renin-mediated hypertension; rupture only >3 cm. Treat if rupture, growth, pregnancy, **>2 cm**, symptoms.

- Congenital AVF (3rd–4th decade): bruit ≈75%, haematuria >75%, hypertension and high-output failure ≈50%; rarely resolves. Acquired may close within months → observe.

8 · Collecting system anomalies

- P1 Pelvicalyceal anomalies other than PUJ obstruction — *calyceal diverticulum (types I/II, stones ≈40%, delayed contrast pooling, percutaneous ablation + fulguration), hydrocalycosis, megacalycosis (non-obstructive, 12–20 calyces, male 6:1, no surgery), infundibulopelvic stenosis (bilateral, >⅓ renal failure), extrarenal and bifid pelvis (≈10%)*
- Megacalycosis bilateral in males; mild concentrating defect only.
- Extrarenal pelvis: confirm with diuretic renography before calling it obstruction.
- Infundibulopelvic stenosis: yearly creatinine, eGFR, urinalysis; falling function → ACE inhibitor or calcium-channel blocker, protein restriction; surgery planned on 3-D MRU.

> **KEY POINT**
> Collecting-system anomalies split on one question — is there obstruction? Dilatation without obstruction (bifid, extrarenal pelvis, megacalycosis) needs nothing; stasis (diverticulum, hydrocalycosis) is treated for symptoms; infundibulopelvic stenosis threatens function through bilateral dysplasia.

9 · Landmark trials

Trial	Design	Result	Take-home
RAFT (Renal Anhydramnios Fetal Therapy)	Non-randomised multicentre; isolated BRA, serial amnioinfusion before 26 wk	94% live births; 82% survived ≥14 d with dialysis access; only 35% discharged on peritoneal dialysis; stopped early	Prevents lethal lung hypoplasia but leaves neonatal ESRD — research only

50 Renal Dysgenesis and Cystic Disease of the Kidney

1 · Genetics in one table

Gene	Renal phenotype
PAX2	Renal hypodysplasia or aplasia
EYA1	Branchio-oto-renal (BOR) syndrome
HNF1B	Renal cysts and diabetes syndrome
PKD1, PKD2 (primary cilium)	Polycystic kidney disease

2 · Dysgenesis: dysplasia vs hypoplasia

- **Dysgenesis** = maldevelopment of size, shape or structure: dysplasia and hypoplasia. Agenesis → *Ch 49*.
- **Dysplasia** is **histological**: arrested metanephric differentiation from a wolffian duct–UB–blastema error. Sine qua non: **primitive ducts** with fibromuscular collars; also cartilage, immature glomeruli. Not all cysts are dysplastic, not all dysplasia is cystic.
- Cause: primary UB failure (Fraser, BOR, renal–coloboma, Kallmann, Townes–Brocks, HDR) or fetal outflow obstruction. **Leading cause of childhood ESRD.** US: small, echogenic (brighter than liver), lost corticomedullary differentiation ± cysts.
- **Hypoplasia** = reduced mass from too few cells; simple hypoplasia has normal nephron number.

Entity	Defining feature	Course / management
Oligomeganephronia	Few, hypertrophied nephrons; usually bilateral	After year 1: polyuria, polydipsia, short stature; **proteinuria first sign**; rapid fall in teens → ESRD second decade. Fluids, correct salt loss/acidosis, protein 1.5 g/kg/d, ACE inhibitor; living related donor acceptable (not familial)
Ask-Upmark (segmental hypoplasia)	Mostly acquired reflux nephropathy; deep "slit-scars", thyroid-like tubules	Girls ≥10 y, **severe hypertension**; partial/total nephrectomy cures BP regardless of renin (persisting BP → contralateral scar); bilateral → medical, dialysis/transplant. Reflux correction does not cure hypertension
Hypodysplasia	With ectopic orifice, megaureter, ureterocele, PUV, prune-belly	**Dysplasia ∝ degree of ureteric ectopia**: lateral → reflux-associated; medial/ureterocele → obstruction-associated. PUV: mild (subcapsular cysts) vs severe (cysts, cartilage). Prune-belly: tortuous ureters, usually no outlet obstruction

KEY POINT

The ureteric orifice predicts the kidney: a bud from the wrong place on the wolffian duct meets the wrong blastema, so ectopia, obstruction and dysplasia travel together.

3 · Cystic disease: concepts and classification

- Cysts arise from tubules by **epithelial proliferation** ("second hit"), **fluid secretion** (net NaCl secretion) and **abnormal extracellular matrix**; the **primary cilium** is central.

- **Multicystic** (MCDK) = dysplastic, no normal nephrons; **polycystic** = normal kidney whose nephrons are progressively replaced by cysts.

Inheritable	Non-heritable
ARPKD	MCDK
ADPKD	Benign multilocular cyst (cystic nephroma)
Juvenile nephronophthisis (recessive) / medullary cystic disease (dominant)	Simple cysts
Congenital nephrosis; familial hypoplastic glomerulocystic disease	Medullary sponge kidney; sporadic glomerulocystic disease
Syndromes: tuberous sclerosis, VHL	Acquired renal cystic disease; calyceal diverticulum

Extrarenal clue decides: hepatic fibrosis → ARPKD; berry aneurysm → ADPKD; retinitis pigmentosa → nephronophthisis; angiofibroma → TSC; haemangioblastoma → VHL. Chromosome 16 carries PKD1 and TSC2.

4 · ARPKD

- P3 Autosomal recessive polycystic kidney disease. — *PKHD1 chr 6, fibrocystin; always congenital hepatic fibrosis; oligohydramnios, pulmonary hypoplasia, 30–50% neonatal death; homogeneously hyperechoic kidneys; scan parents; supportive care; ARegPKD early bilateral nephrectomy warning*
- ≈½ of neonatal survivors alive at 10 y; later-onset disease: hepatic problems dominate (portal hypertension with normal liver enzymes). Pyramids merge with cortex on US. CT if tumour suspected (Wilms, mesoblastic nephroma).
- Nephrectomy risks: nephron loss (earlier dialysis), hypotension after bilateral removal (also predicted neurological injury). Remaining kidney may grow fast after unilateral nephrectomy → second operation; buy time.

5 · ADPKD — paediatric focus

- P2 ADPKD (Adult polycystic kidney disease) — *PKD1 (16, 85%, ESRD 50s) vs PKD2 (4, 15%, ESRD 70s); pain, haematuria ≈50%, stones 20–30%, hypertension ≈50% at 20–35 y; liver cysts, aneurysm 10–30%; Ravine criteria; BP <130/80 with ACEi/ARB, tolvaptan (TEMPO, REPRISE), lipophilic antibiotics, unroofing*
- ≈10% de novo; in utero/infantile onset is aggressive. Children: cysts most by 10 y (PKD1); ≈17% look unilateral/asymmetric.
- **Contiguous gene syndrome**: deletion of PKD1 + TSC2 → TSC with severe early polycystic kidneys.
- **Hypertension with normal function is the key paediatric finding**; tight control slows decline, cardiac and aneurysm morbidity. Early ESRD risk: PKD1, male, haematuria <30 y, hypertension <35 y, hyperlipidaemia, sickle trait.
- Liver cysts rare in children, worse in females. RCC not increased.
- Negative US young does **not** exclude PKD2 (sensitivity ≈⅔ <30 y) → genetic testing when certainty matters; counsel first. **No family history**: bilateral cysts + ≥2 of bilateral enlargement, ≥3 hepatic cysts, cerebral aneurysm, arachnoid/pineal/pancreas/spleen cyst.
- CT haemorrhagic cyst 50–90 HU; MRI bright T1/dark T2. Tolvaptan approved 2018 (aquaresis, hepatotoxicity). Stones: uric acid and oxalate equally; cysts camouflage

hydronephrosis.

- Aneurysm screening (MRA) only for family history, prior rupture, high-risk job, major surgery, anxiety; <7 mm observe; coil/clip >7–10 mm or symptomatic.

> **KEY POINT**
> ADPKD is a volume disease: cyst growth stretches vessels (hypertension), compresses nephrons and hides stones and tumours. Every therapy is judged by whether it slows total kidney volume.

6 · Juvenile nephronophthisis (NPH) / medullary cystic kidney disease (MCKD)

- Ciliopathies: **tubular basement membrane thickening, tubular atrophy with corticomedullary cysts, interstitial fibrosis**; small-to-normal kidneys. **Polyuria/polydipsia >80%** (vasopressin-resistant, **salt wasting**); ESRD 5–10 y after presentation.

	Juvenile NPH	MCKD
Inheritance	Autosomal recessive	Autosomal dominant
ESRD	Mean 13 y; **10–20% of childhood renal failure**	Third–fourth decade
Hypertension	Absent	Present
Extrarenal	20%: **retinitis pigmentosa (Senior-Løken)**, Joubert, Cogan apraxia, hepatic fibrosis, Bardet–Biedl	None; type 2 — hyperuricaemia, gout in teenage males

- US often misses cysts; CT/MRI better. Bardet–Biedl = obesity, intellectual disability, polydactyly, retinitis pigmentosa, hypogenitalism.
- **Sodium replacement; avoid salt restriction and diuretics**; transplant — no recurrence in grafts.

7 · Syndromes with renal cysts

Tuberous sclerosis complex (TSC)

- AD; **TSC1 (chr 9), TSC2 (chr 16)**. Epilepsy ≈80%, intellectual disability ≈60%, adenoma sebaceum ≈75%; earliest skin sign ash-leaf macule. Diagnosis: 2 major or 1 major + 2 minor (AML major; multiple renal cysts minor).
- Kidney: cysts (mimic ADPKD), **AML 40–80%** (bilateral, multiple, after 10 y; never metastasise; risk = haemorrhage), RCC ≈2% (younger, female, bilateral — suspect in growing fat-free/calcified lesion).
- At diagnosis: abdominal MRI, BP, GFR. AML fat ≈−50 HU on CT; **MRI preferred** (25–30% fat-poor AMLs missed on US); recognise fat-poor AML to avoid surgery.
- **AML >3 cm and growing → everolimus (mTOR inhibitor) first line**; acute bleed → embolisation then corticosteroids; kidney-sparing surgery second line; avoid nephrectomy. Imaging every 1–3 y (yearly if near 3 cm/growing); BP, proteinuria, GFR yearly; ACEi/ARB for hypertension.

Von Hippel–Lindau disease

- P2 VHL gene and von Hippel–Lindau disease — *AD, 3p25, pVHL degrades HIF → VEGF; lesion frequencies (cysts 76%, clear cell RCC, phaeo, haemangioblastoma, epididymal cystadenoma); diagnostic criteria; VHL Alliance 2020 surveillance (abdominal MRI 2-yearly from 15); nephron-sparing surgery, recurrence 75–85%, DSS 81–94%; partial adrenalectomy*

- Surveillance also: exam yearly from 1 y, BP from 2; stop routine imaging at 65; full check before conception. Bilateral radical nephrectomy only for high-grade disease.

8 · Multicystic dysplastic kidney (MCDK)

- Non-communicating cysts, **no normal parenchyma, no function**, atretic ureter ("bunch of grapes"). Types: infundibulopelvic (commoner) and hydronephrotic.
- **Commonest renal cystic disease of the newborn; second commonest neonatal abdominal mass**; 1 in 1000–4000; male 55–70%. Bilateral = lethal. Obstruction theory vs **UB-gene theory** (EYA1, PAX2, WT1, GDNF; favoured — explains contralateral anomalies). Rarely in a duplex upper pole or horseshoe moiety.
- **Contralateral**: hypertrophy; **VUR 18–43%, PUJ obstruction 3–12%**.
- ≈40% involute (later "agenesis"); hypertension and Wilms not increased.

US feature	MCDK	Severe PUJ obstruction
Cysts	Haphazard, varied, non-communicating	Calyces around central pelvis, communicating
Central large cyst	Absent	Present (pelvis)
Isotope uptake	None	Usually some

ALGORITHM

Prenatal MCDK → US day 2–3 and at 1 month (contralateral kidney, bladder) → DMSA or MAG3 if doubt → VCUG only if UTI or abnormal contralateral kidney/bladder → primary care: BP, mass, UTI

- **Nephrectomy** only for mass effect, hypertension, cyst rupture or **increasing solid tissue**. Routine US surveillance (3–12-monthly) of no proven benefit. Malignant transformation extremely low, perhaps nil.

KEY POINT

MCDK removes itself; management looks to the solitary contralateral kidney and its reflux or obstruction. Mistaking a non-functioning PUJ kidney for MCDK costs little.

9 · Benign multilocular cyst (cystic nephroma)

- Spectrum (no conversion): multilocular cyst → cystic partially differentiated Wilms (CPDN) → cyst with Wilms nodules → cystic Wilms. Adult counterpart: multilocular cystic RCC.
- **Bimodal: boys <4 y (M 2:1, flank mass) and women >30 y (F 8:1, haematuria from herniation into pelvis)**.
- Imaging separates it from MCDK but not from cystic Wilms, mesoblastic nephroma, clear cell sarcoma → **excision is diagnosis and treatment** (nephrectomy in children; partial if localised). Wilms nodules → NWTS staging.

10 · Simple cysts in children and Bosniak

- **US criteria (all four)**: anechoic; thin smooth wall; posterior enhancement; round/oval. CT: −10 to +20 HU, no enhancement. Hyperdense (20–90 HU) cyst benign if non-enhancing, homogeneous, ≤3 cm and ≥¼ wall extrarenal. Calcification: 2% of cysts (peripheral) vs 10% of RCC (central).
- Puncture now rare: suspected infection, discordant imaging, borderline lesion in poor surgical candidate.

- **Children**: simple cyst <15 mm, asymptomatic → repeat US at 2 y, discharge if stable. Continue if ≥15 mm, complex, family history, or GU anomalies. Bilateral cysts without liver cysts — do not label ADPKD without family history/genetics.

Bosniak	Features	Management
I	Simple	None
II	Fine septa, minimal calcification, hyperdense ≤3 cm	None
IIF	More septa/calcification, no measurable enhancement	Follow-up (≈11% progress)
III	Thick/irregular enhancing walls or septa	Excision (paediatric III–IV: >90% intermediate/malignant)
IV	Enhancing solid components	Excision

Symptomatic benign cyst: aspiration ± ethanol sclerosis or laparoscopic unroofing; infected → drainage + antibiotics. Bosniak 2019 adds MRI and down-classifies to curb over-treatment.

11 · Medullary sponge kidney, glomerulocystic and acquired cystic disease

- P2 Medullary sponge kidney — *ectatic collecting ducts, urographic triad ("brush bristles", "bouquet of flowers"), presents >20 y, colic 50–60%; hypercalciuria ⅓–½, incomplete dRTA 30–40%; potassium citrate first, thiazide, phosphate (not with urease infection)*
- **Sporadic glomerulocystic disease**: bilateral enlarged kidneys, Bowman-space cysts; diagnosis of exclusion.
- P2 Acquired renal cystic disease — *≥3 cysts/kidney in ESRD; uraemia not dialysis; 10% pre-dialysis → >90% at 10 y; small kidneys; haemorrhage, adenomas, RCC (aggressive in native kidneys after transplant); bleeding → embolisation/nephrectomy; tumour >3 cm excise, <3 cm nephrectomy or yearly CT*
- ARCD in children on dialysis 20–45%. Retroperitoneal bleed → nephrectomy (occult RCC likely); heparin haematuria → peritoneal dialysis. Avoid iodinated contrast pre-dialysis.
- **Parapelvic/renal sinus cysts** (lymphatic) mimic hydronephrosis on US — delayed contrast CT shows undilated displaced system; conservative.

12 · Landmark trials

Trial	Design	Result	Take-home
HALT-PKD	Randomised: lisinopril + telmisartan vs lisinopril; rigorous vs standard BP	Dual blockade no benefit; low BP slowed TKV growth, ↓ LV mass and albuminuria, GFR unchanged	Single RAS blocker; tight BP early
TEMPO / REPRISE	Randomised, tolvaptan vs placebo	Slower TKV growth and GFR decline	First disease-modifying drug; aquaresis, hepatotoxicity
ARegPKD registry	Observational ARPKD cohort	Bilateral nephrectomy <3 months → worst neurological outcome	Avoid early bilateral nephrectomy

51 Pathophysiology of Urinary Tract Obstruction

1 · Definition and classification

- **Obstruction**: any anatomic or functional impediment to flow, calyces to meatus, that injures kidney or lower tract. **Acute/chronic · partial/complete · intrinsic/extrinsic · unilateral/bilateral**.
- Defined by effect on **flow and function, not dilation**: partial vs high-grade, not the image, decides treatment.
- ≈ **1 in 500** live births; high-grade obstruction lifetime ESRD risk up to **30%**.

	LUTO	Upper tract
Commonest	PUV and urethral atresia ≈ 50%	UPJO, 1 in 1500; M:F 2:1
Others	Anterior urethral valves, megalourethra, prolapsing ureterocele, cloaca, megacystis–microcolon–intestinal hypoperistalsis, trisomy 18/21	UVJO, ectopic ureter, ureterocele, fibroepithelial polyp, stones, crossing vessel
Outcome driver	ESRD + pulmonary hypoplasia	Unilateral function loss; contralateral kidney buffers

Presentation (site, severity, age, duration): prenatal hydronephrosis (most at 18–20 wk); postnatally mass, fever/recurrent UTI, flank pain, polyuria or retention, failure to thrive, electrolyte imbalance, respiratory failure from pulmonary hypoplasia.

> **KEY POINT**
> Trap: **detrusor hypertrophy** is a lower-tract cause of upper-tract (UVJ) obstruction — the thick wall narrows the intramural ureter. Neurogenic bladder obstructs without an anatomical lesion.

2 · Haemodynamic changes

Urine production continues; pressure rises → net filtration pressure, GFR and RBF fall → ischaemia; injury begins within **2 h** of complete obstruction; bilateral/solitary kidney → more profound.

† Phase (complete)	Unilateral	Bilateral / solitary
I (0–1.5 h)	RBF ↑, ureteric pressure ↑ (afferent dilatation: PGE2, NO)	Modest RBF ↑
II (1.5–5 h)	RBF ↓, pressure still ↑ (efferent constriction)	RBF ↓, pressure ↑
III (> 5 h)	RBF ↓, pressure ↓ (afferent constriction: angiotensin II, thromboxane)	Pressure stays high (retained solute, ANP) — basis of POD

- **Hypertension** (RAAS): bilateral ≈ **77%**; unilateral ≈ **20%** (renin lateralises).

3 · Tubular function

- **AQP1–4 and Na transporters downregulated** → concentrating defect, salt wasting, polyuria; **distal RTA**.
- Early: wastes Na, K, HCO_3, Mg, Ca, phosphate. Prolonged: retains K, H^+, Cl, ammonium — K **wasted early, retained late**.

- FENa < 1% in acute obstruction; > 1% with tubular injury.

4 · Pathology and fibrosis

- Immature kidney: apoptosis, impaired nephrogenesis — nephron number permanently capped. Mature: tubular proliferation → collagen → scarring.
- Fibrosis (↑ α-SMA, ECM) = final common path. Stretch mechanotransduction (initiator) → TNF-α/interleukins → **TGF-β → Smad** (angiotensin II amplifies); **miR-34a** → epithelial–mesenchymal transition.
- **Bladder**: hypertrophy → compensation → decompensation; IL-6, TNF-α, MCP-1 → poor compliance. Storage pressure > **40 cm H_2O** injures kidneys.

5 · Recovery and compensatory growth

- Relief can reverse damage but **incomplete recovery is common**; obstruction erodes renal reserve (acute-on-CKD → rapid AKI).
- Compensatory growth (IGF-1, TGF-β1, HGF) = **glomerulomegaly, not new nephrons**.

6 · Post-obstructive diuresis (POD)

- P1 Post-obstructive diuresis — *unilateral ≈ **2%**, bilateral ≈ **50%**; aquaporin/transporter loss, medullary washout, retained solute; risk after valve ablation (oligohydramnios, preterm, creatinine); monitoring, partial replacement, avoid overcorrection*
- Preterm infants: low AQP2 → POD risk.

> **ALGORITHM**
> Monitor intake/output, serum and urine electrolytes → replace losses, correct electrolytes → **do not over-replace** (perpetuates POD) → usually resolves in **24 h**, seldom > 48 h

7 · Fetus and child

- P3 Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) — *UTD APD thresholds (**≥ 7 mm** at 16–27 wk, **≥ 10 mm** ≥ 28 wk), serial fetal urine (Na < 100, Cl < 90, osmolality < 200, β2-microglobulin < 6), shunt/fetal cystoscopy, amnioinfusion*
- Kidneys seen from 9 wk, bladder by 12 wk (not seen by 15 wk = abnormal); visible fetal ureter = dilated. Megacystis: bladder failing to empty over 40 min at ≥ 18 wk.
- Second-trimester oligo/anhydramnios → mortality **60–80%**. Favourable fetal urine also Ca < 8 mg/dL; last sample reflects current function.
- Postnatal: US; MAG3 drainage; DMSA scar; VCUG outlet/VUR. Whitaker obstructed if pelvic pressure > **12 cm H_2O** or continuous rise. Cystatin C GFR most accurate in children.

8 · Treatment principles

Temporise, then define: drain to buy function, reconstruct later; manage POD and electrolytes carefully.

Option	Indication / key point
CIC	Neurogenic bladder; ↓ UTI, stones, retention
Overnight catheter drainage	Valve bladder with nocturnal polyuria and progressive CKD
Vesicostomy	Infant with complex LUTO, unstable or unable to CIC
Percutaneous nephrostomy	Upper tract; needs some hydronephrosis

Option	Indication / key point
Stent / cutaneous ureterostomy	Temporising; stent excellent for extrinsic obstruction
Pyeloplasty / reimplantation	Definitive; anastomose anterior to aberrant vessel; non-refluxing reimplant in children

- NSAIDs only if renal function preserved. Oliguria/AKI: treat cause, optimise fluids and perfusion; CKD: BP, diet. ESRD → dialysis/transplant → *Ch 61*.

52 Ectopic Ureter, Ureterocele and Ureteral Anomalies

1 · Definitions and classification

- Ectopic ureter and ureterocele: one defect of **ureterotrigonal connection** on a continuum (intermediate cases exist); appearance changes technique, not strategy.
- **Ectopic ureter**: single or duplex ureter not entering the trigone; ectopia = entry **at or distal to the bladder neck** (merely near it is not ectopic).
- **Ureterocele**: **cystic dilation of the distal ureter** within the bladder or spanning bladder neck and urethra; never entirely urethral, never attached to wolffian structures.
- In a **duplex system, both lesions always involve the upper pole** ureter.

	Ectopic ureter	Ureterocele
Distal ureter	Dilated or not, opens ectopically	Cystic dilation
Position	At/distal to bladder neck; perineum, vagina, rectum (F) or wolffian ducts (M)	Bladder or spanning bladder neck and urethra
Wolffian attachment	Yes (male)	Never
Incontinence (untreated)	Yes, in females	No

Ureterocele classification

- P3 Ureterocele — *Glassberg intravesical vs ectopic (most useful, predicts TUI outcome); Stephens incl. cecoureterocele; Churchill functional*
- Churchill: upper pole only → whole ipsilateral kidney → contralateral too (reflux or bladder outlet obstruction (BOO)), last group ~1/4; outcome prediction unreliable.
- **Cecoureterocele**: large incompetent intravesical orifice; cavity tunnels submucosally beyond bladder neck into urethra. Often missed preoperatively; troublesome at incision; incontinence risk.
- Say "single intravesical" not "simple" ureterocele — it can still obstruct severely.
- Distinguish ectopic ureterocele from an intravesical one that **prolapses** into the urethra only on voiding.
- **Ureterocele disproportion** (non-obstructive): small dysplastic non-dilated upper pole invisible on imaging, no ureteric dilation — typical ureterocele, apparently normal kidney.

2 · Embryology essentials

- Duplex: upper pole bud arises **later and more cephalad** on the wolffian duct → incorporated later → orifice carried **caudally** (ectopic).
- **Weigert–Meyer rule**: **upper pole orifice caudal (and medial) to lower pole orifice**. Upper pole → ectopia/ureterocele (obstruction); lower pole → short tunnel → vesicoureteral reflux (VUR).
- **Chwalle membrane** failing to rupture → ureterocele (classic theory).
- Classic: common nephric duct forms trigone. Revised: it regresses by apoptosis; trigone from endoderm.
- Renal dysplasia is **downstream** of the connection defect, not parallel.
- Ectopic ureter never enters müllerian structures directly; reaches vagina/cervix/uterus via the **Gartner duct** — a cystic Gartner duct rupturing into the vagina underlies female

incontinence.

3 · Ectopic ureter — sites and sex asymmetry

- P3 Embryology and management of ectopic ureter. — *female vs male sites, continuous dribbling vs never incontinent, epididymitis, Zinner syndrome, Gartner duct*

> **KEY POINT**
> Why does only the girl leak? The female ureter may open distal to the external sphincter and leaks continuously; the male ureter, confined to wolffian derivatives above it, presents as infection or obstruction.

- Male sites: bladder neck, prostatic urethra, vas, seminal vesicle, ejaculatory duct. Zinner: retrograde injection shows ureter–seminal vesicle–vas communication.
- **Single-system ectopic ureter** may mimic an "absent kidney" (small poorly functioning unit missed on ultrasound) → contrast CT or renal function study.
- Distal obstruction may cause severe hydroureteronephrosis, sometimes non-function.
- **Bilateral single-system ectopia** (rare): hypoplastic bladder, poor trigone and bladder neck, bilateral dysplasia.

4 · Clinical presentation

Prenatal

- Most found as prenatal hydronephrosis; specific diagnosis postnatal. Duplex hard to see unless one moiety dilated. Fetal upper pole "cyst" = upper pole hydronephrosis until proven otherwise.
- **Diagnosis is made at the bladder** (upper tracts identical). A large ureterocele can fill and mask the bladder.
- Record upper pole thickness/echogenicity; ipsilateral lower pole or contralateral dilation suggests reflux. Ureterocele BOO rarely causes oligohydramnios.
- Whole kidney-and-ureter dilation suggests single system; may mimic megaureter or severe reflux until after birth.

Incidental and infection

- "Presumed ovarian cyst" may be a dilated ureter — include in differential of cystic pelvic mass.
- Urosepsis → ultrasound in all, even after a normal late prenatal scan. Unresponsive sepsis → **urgent transurethral incision (TUI)** of ureterocele, or percutaneous/open drainage (usually end ureterostomy) of ectopic ureter.
- Ectopic ureter: subacute low-grade fever with spikes; **negative culture does not exclude infection**. Girls: purulent perineal discharge → ultrasound. Unobstructed ectopic ureter rarely infects.
- **Any young male with bacterial epididymitis → upper tract ultrasound** (often recurrent before diagnosis).

Incontinence

- Toilet-trained girl, verified **continuous low-volume dampness day and night** with normal voiding → evaluate. Untreated ureteroceles do not cause incontinence.

- Key question: **can she be dry for 30–60 minutes?** Ectopic ureter — no (but parents of other wetters also say "always wet").
- Pre-training: dribbling at changing. Rarely intermittent (leak through Gartner duct membrane).
- Leaking moiety may be non-dilated and missed on imaging — suspicion is history-driven; exclude voiding postponement, posturing and constipation.

Ureterocele prolapse

- **Smooth, congested, mucosa-covered interlabial mass** from the urethra, often with voiding difficulty; lobulated grape-like mass → **sarcoma botryoides**.
- Bladder ultrasound confirms; VCUG shows prolapse, lower pole reflux, "drooping lily".

Late presentation

- Teen/adult: infection or pain, rarely incontinence. Adult single-system non-obstructing ureterocele often contains a **stone**.
- Moiety usually has little function → upper pole heminephrectomy (mainstay) or upper-to-lower ureteroureterostomy.

5 · Evaluation

- Examine perineum in every continuously wet girl: ectopic orifice **between urethral meatus and vagina**, cannulate with angiocatheter for contrast. Dilated upper pole may be palpable in an infant.

Imaging

Test	Answers	Note
Ultrasound	Anatomy; inferred function; bladder	First always; cannot separate ectopic ureter from ureterocele above bladder
Voiding cystourethrogram (VCUG)	Reflux (lower pole, contralateral, into ureterocele); bladder base, urethra	Optional if obstruction alone dictates incision; repeat after incision
Dimercaptosuccinic acid (DMSA)	Differential function — gold standard	Only "some vs none"
Mercaptoacetyltriglycine (MAG3) renogram	Function + drainage	Observation candidates; after 2 months
MR urography	Most detailed anatomy + function	Complex anatomy, non-dilated upper pole; needs sedation
Intravenous pyelogram (IVP)	Anatomy + qualitative function	Non-dilated ectopic ureter with incontinence

- **Ureterocele**: thin-walled cyst within bladder. **Pseudoureterocele**: dilated ectopic ureter indenting bladder — **bilaminar, thicker wall, lumen outside bladder**.
- Pitfalls: large ureterocele fills bladder; full bladder effaces it (mimics ectopic ureter); lobulated one looks like two cysts; cecoureterocele extension hard to see. Observe over fill and void (also gives emptying and wall thickness).
- VCUG: lower pole reflux recognised by calyceal orientation; **"drooping lily"** = lower pole displaced down/out by dilated upper pole. With BOO or severe bilateral obstruction, puncture

on ultrasound alone, then post-puncture VCUG.

- † Excretory urography of single-system intravesical ureterocele: **"cobra head" (spring-onion)** — dilated distal ureter with radiolucent halo.
- No criteria for "normal" upper pole function; **preoperative function does not predict function after decompression**.

VCUG terms

Term	Meaning	Significance
Effacement	Flattens with filling	Mimics ectopic ureter
Prolapse	Moves into urethra	Outlet obstruction; predicts secondary surgery
Eversion	Diverticulum-like protrusion of bladder base	**Weak trigonal floor → likely open repair**

Endoscopy

- **Site selection** before incision: start with little filling — lowest part of ureterocele (best incision site) seen then.
- Document urethra, bladder neck, trigone, orifices; detect urethral extension (cecoureterocele) so incision leaves no obstructing segment.
- Retrograde injection: confirms ureterocele disproportion or ureter–vas–seminal vesicle communication.
- Ectopic orifice at bladder neck often patulous. Ectopic ureter separated from urethra by a thin layer mimics ureterocele, goes with bladder neck defects → wider excision with bladder neck reconstruction.
- Patulous bladder neck / cecoureterocele → later incontinence risk; consider bladder neck tightening (videourodynamics may help).

6 · Management principles

- Goals: **preserve renal function; eliminate infection, obstruction and reflux; maintain continence; minimise total morbidity**.
- Early daily antibiotic prophylaxis in significant hydronephrosis, even without reflux.
- Relieving obstruction may **create** reflux or let lower pole reflux resolve — the central trade-off.
- Function unpredictable before/after decompression; recovery limited. Decisions empirical.
- **Fetal intervention**: none for ectopic ureter; ureterocele only if large and obstructing all units (aspiration/laser — experimental).

Neonatal algorithm

> **ALGORITHM**
>
> Prenatal hydronephrosis → postnatal ultrasound → prophylaxis if significant hydroureteronephrosis → VCUG (optional if it won't change decompression) → renal scan after 2 months
>
> Small ureterocele, minimal hydronephrosis, no obstruction → observe
>
> Significant obstruction, parenchyma at risk → TUI (or cutaneous distal ureterostomy if endoscopy impossible, until old enough for reimplant)
>
> Unresponsive sepsis → TUI (ureterocele) / nephrostomy or end ureterostomy (ectopic ureter); elective drainage still needed later

- Optimal age for elective decompression undefined.
- Infant ectopic ureter: side-to-side non-dismembered refluxing anastomosis to bladder, intended definitive — no long-term data.
- **Observation** (selected ureteroceles): no obstruction of ipsilateral lower pole or contralateral kidney, lower pole reflux ≤ grade III; non-functioning upper pole (e.g. multicystic dysplastic, no ureteric dilation); or no obstruction on renogram. Dilation and reflux may resolve; counsel on unpredictable acute presentation.

7 · Operative options

Transurethral incision (TUI) of ureterocele

- Relieves obstruction in **78–97%**. Choice in **sepsis** and the **infant**.
- **Low transverse full-thickness incision**, cutting current, **as distal and close to bladder floor as possible** (less reflux). Angled-tip wire, Collins knife or laser; incise deeply. Adequacy: urine jet or urothelium seen inside.
- Ectopic ureterocele: longitudinal incision into urethral portion, or separate intravesical and urethral punctures (leave no obstructing urethral flap).
- **"Watering-can"**: multiple holmium punctures — equal decompression, less de novo reflux.
- Well child: no catheter, outpatient. Ultrasound + VCUG at **4–6 weeks**.
- New reflux **0–50%**. **Intravesical**: decompression without reflux 70–80%. **Extravesical**: secondary surgery up to 70%. Reoperation RR 1.74 with lower pole reflux, 2.78 if extravesical (not additive). Single-system better than duplex.
- Persisting reflux → usually ureterocele excision and reimplantation (some resolve).

> **KEY POINT**
> Why TUI first even if half need more surgery? Outpatient, may be definitive, defers the second operation to an older child, and decompression often avoids tapering at reimplant. Less appealing in an older child with a massive non-functioning upper pole — definitive surgery at diagnosis.

Upper pole partial nephrectomy (heminephrectomy)

- For non-functioning upper pole or doubtful drainage from massive dilation. Open (flank or dorsal lumbotomy), laparoscopic or robotic. Primary concern: **do not damage the viable lower pole**.
- Key steps: gentle retraction → upper pole ureter as handle passed **behind the renal vessels** → ligate upper pole vessels (usually 2–3) → resect at demarcation → close over fat pedicle. Mannitol with pedicle clamp; papaverine for vasospasm. Laparoscopic: ties avoid clip avulsion; IV methylene blue checks leak.
- **Dissect on the wall of the upper pole ureter**; leave its back wall on the lower pole ureter near the bladder — shared blood supply.
- Distal stump: small → ligate and leave; large or obstructed-and-refluxing → excise low without injuring bladder neck/vas.
- Postoperative Doppler ultrasound (urinoma, lower pole flow). Clipping ureter alone is tentative (pyonephrosis reported).
- Outcomes: ectopic ureter — usually curative. Ureterocele — lower pole reflux resolves in up to 20%; new reflux **15–50%**; secondary surgery **40–50%**.

Ureteroureterostomy / pyeloureterostomy

- Upper pole end-to-side into lower pole system; functioning and, increasingly, non-functioning upper poles (hypertension concern unsubstantiated).
- **Distal** (inguinal): **stent lower pole ureter cystoscopically first**. **Proximal** preferred for dilated upper pole/size mismatch.
- Minimal medial dissection; resect upper pole ureter low on its wall; stent recipient only, never across the anastomosis. Ureterocele stump not excised if well decompressed.
- Robotic-assisted (RALUU) ≈ open, slightly shorter stay; size mismatch no barrier.

Lower tract reconstruction

- **Ureterocele excision + common sheath reimplantation**: relieves obstruction and reflux in one; fallback for persistent reflux. Risks: bladder neck, sphincter, vagina.
- Mobilise **both ureters as a unit** (never separate — shared blood supply) → taper upper pole ureter if needed → excise ureterocele to bladder neck → **plicate detrusor defect** → submucosal reimplant. Below bladder neck protect sphincter; fulgurate unexcised remnant.
- Cecoureterocele remnant can act as obstructing **urethral windsock** flap-valve; watch for voiding dysfunction.
- **Marsupialisation** (excise roof, no back-wall dissection): small ureteroceles without detrusor defect; complete excision preferred.
- Persisting reflux **5–10%**, more with tapering.
- **Total reconstruction** (heminephrectomy + excision + lower pole reimplant): **older child, massive ureterocele, non-functioning upper pole, significant lower pole reflux**.

End ureterostomy for ectopic ureter

- Neonate with sepsis or massive dilation; decompresses, function assessed later.
- **Do not trim redundant ureter** (shortens; length needed for reimplant). Stoma at lateral end of future Pfannenstiel.
- DMSA at ~4 months or after 6 months → salvageable → reimplant/ureteroureterostomy; none → nephrectomy.

8 · Decision summary

Ectopic ureter (duplex)

> **ALGORITHM**
>
> Keep upper pole + lower pole reflux → common sheath reimplant, or lower pole reimplant + distal ureteroureterostomy
>
> Keep + no lower pole reflux → proximal or distal ureteroureterostomy
>
> Function unclear / massive dilation / sepsis → temporary end ureterostomy → DMSA → reconstruct or remove
>
> Remove (marked dilation, preference) → upper pole nephrectomy with ureterectomy

- If lower tract surgery is needed anyway for reflux, upper tract surgery may be avoided.
- Single-system: preserve vs remove by function and preference — no outcome data either way.

Ureterocele (Table 52.1)

Procedure	Ideal indication	Advantage	Limitation
TUI	Infant; large ureterocele with VUR	Outpatient; effective; occasionally definitive	Reflux into ureterocele moiety → bladder surgery
Upper pole nephrectomy	Older; large non-functioning upper pole; no VUR	Removes pathology; avoids bladder	Lower pole risk; may still need bladder surgery
Ureteroureterostomy / pyelostomy	Older; functioning upper pole; no VUR	Drains with little risk	Ureterocele left; may reflux
Excision + common sheath reimplant	VUR; functioning upper pole without major dilation	Removes obstruction and reflux; no renal risk	Complex; vagina/bladder neck risk; tapering

KEY POINT

VUR is the pivot: no VUR → upper tract solution; VUR → bladder-level solution — except in the infant, where TUI defers bladder surgery. Upper pole function then picks nephrectomy vs ureteroureterostomy. TUI failure costs little and may simplify the next operation.

9 · Complications

- **Loss of lower pole function** after heminephrectomy (most significant; may be recognised late) — prevent by preserving inter-ureteric blood supply and gentle traction.
- **Urinoma**: up to 20% after laparoscopic/robotic cases, mostly when defect not closed; asymptomatic → observe.
- **De novo reflux** after TUI or heminephrectomy → observe if no infection; else excision and reimplantation.
- **Voiding dysfunction**: infrequent voiding common, incontinence uncommon. Inadequate trigonal support → ballooning trigone behind bladder neck → obstruction, poor emptying, infection, recurrent VUR (also after TUI of cecoureterocele).
- Assess with VCUG **including lateral voiding views** ± videourodynamics; treat by trigonal/bladder neck reconstruction, clean intermittent catheterisation (CIC) or bulking.

10 · Other ureteral anomalies

Duplication

- ~**1 in 125** (0.8%) at autopsy; F:M 1.6:1; unilateral 6× bilateral. **Look for contralateral duplication** — an occult second ectopic ureter explains persistent wetting.
- **Bifid pelvis** = single ureter; **bifid ureter** = partial duplication joining below the UPJ.
- **Lower pole gets single-system disease** (UPJ obstruction, VUR); **upper pole gets ureteral-formation disease** (ectopia, ureterocele).
- Lower pole UPJ obstruction: manage as single system; option of pyeloureterostomy to upper pole ureter. Upper pole UPJ obstruction exceptionally rare.
- Lower pole VUR: often high grade, reduced function, slower resolution; common sheath reimplant, endoscopic injection or low ureteroureterostomy.

Triplication

- Smith: type 1 three separate ureters (**35%**, commonest); 2 incomplete, two orifices (21%); 3 trifid, one orifice (31%); 4 two ureters, three orifices (13%). Weigert–Meyer usually holds;

violated in quadruplication.

Valves, diverticulum, polyp

- † **Ureteral valves**: transverse mucosal folds with smooth muscle; need proximal dilation, normal ureter below, no other cause; excise segment.
- † **Diverticulum**: usually a **blind-ending bifid ureter**; excise if symptomatic.
- **Fibroepithelial polyp**: commonest at **UPJ**; **89% male, 78% left**; only 22% seen preoperatively (filling defect). At UPJ → pyeloplasty; elsewhere → **ureteroscopic removal**.

Vascular position anomalies

- P1 P2 IVC anomalous development and retrocaval ureter; Retroileal Ureter — *persistent subcardinal vein, right only, type I fishhook vs II, CT urography, transpose anterior; retroiliac at L5–S1 with mesonephric ectopia*
- Retrocaval: always right even with bilateral cavae; mimics UPJ obstruction (pyeloplasty fails).
- Ureteral herniation (inguinal, scrotal, femoral): mostly paraperitoneal, ureteric loop **medial** to the sac.

53 Surgery of the Ureter in Children

1 · Pelviureteric junction (PUJ) obstruction — definition and aetiology

- P1 Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management. — *Koff definition; intrinsic (infant) vs crossing vessel (older, intermittent pain) vs secondary; bilateral 10–40%; associated anomalies up to 50%; SFU grades; MAG3 T½ cut-offs; one-third need surgery*
- Paediatric hydronephrosis: pelvic anteroposterior diameter (APD) **12 mm**; hydronephrosis ≠ obstruction. Koff's definition is retrospective — every indication tries to predict harm before function is lost.
- Crossing vessel: often no intrinsic narrowing after mobilisation, yet **dismembering and transposing the ureter anterior to the vessel prevents recurrence**.
- **Secondary PUJ obstruction** from high-grade vesicoureteral reflux (VUR): avoid simultaneous reimplant + dismembered pyeloplasty (devascularises ureter from both ends) → conservative or **two-stage**.
- **Lower-moiety PUJ obstruction (duplex)**: incomplete duplication → ureteropyelostomy to lower-pole pelvis; complete → standard pyeloplasty; moiety ≤ **5–10%** function → partial nephrectomy.
- Associations: contralateral PUJ obstruction (commonest) → dysplasia → MCDK → agenesis; VUR up to 40% (high resolution); horseshoe kidney, VATER.

2 · Diagnosis and indications

- Antenatal US ≈ 20 weeks; febrile UTI usually responds to antibiotics, few need nephrostomy.
- Neonatal renogram washout confounded by immaturity, hydration, diuretic timing, posture, pelvic compliance — false positives up to **44%**; decide on DRF trend and dilatation, not T½ alone.
- **Pyeloplasty indications**: increasing APD · low or falling DRF · breakthrough infection on prophylaxis · symptoms (pain; failure to thrive/feeding difficulty in infants; haematuria; rarely renin-mediated hypertension).

3 · Anderson–Hynes dismembered pyeloplasty

Gold standard for any access, success ≈ **95%**; originally devised for retrocaval ureter. Preserves a crossing vessel (ureter transposed anterior), excises adynamic segment, allows pelvic reduction.

> **ALGORITHM**
> Expose PUJ → free lower pole (find accessory vessel) → sparing bipolar → stay sutures → decompress pelvis with 21-G needle → dismember ± reduce pelvis → **spatulate laterally** into healthy ureter, stenotic segment as "no-touch" handle → apex to inferior pelvis → tension-free watertight 6-0/5-0 absorbable anastomosis → excise stenotic segment last → irrigate clot → cover with perinephric fat

- Long gap: full renal mobilisation; horizontal lower-pelvis incision. Excess stay-suture tension kinks the ureter on release. Drain usually unnecessary.

Alternative	Best indication	Point
Foley Y-V-plasty	High insertion, **no vessel**	V base on dependent medial pelvis, apex at PUJ
Spiral flap	Long proximal stricture, dependent PUJ	Length:width ≤ 3:1
Vertical flap	Dependent PUJ, large box-shaped extrarenal pelvis	Flap length must match gap
Vascular hitch (Hellström)	Anterior lower-pole vessel, **no intrinsic stenosis**: moderate hydronephrosis, no calyceal dilatation, good cortex and split function; peristalsing PUJ at surgery	Vessel fixed cranially in pelvic wall; no hypertension seen
Uretero-calicostomy	Failed repair, short scarred ureter, dilated lower calyx with thin cortex	Lower-pole nephrectomy exposes calyx → spatulated ureter over stent; capsule not closed near anastomosis

Flaps only for long stenosis where tension-free primary anastomosis is impossible. Non-dismembered (Heineke–Mikulicz) repair inferior.

> **KEY POINT**
> Every variant answers two questions: is there a crossing vessel (dismember), and is the gap too long for tension-free anastomosis (flap, mobilisation)?

> **KEY POINT**
> One fixed principle, many routes: dismember, excise, spatulate, anastomose without tension. Access, stent and robot are choices of convenience; failure is managed by repeating the principle.

4 · Access

- **Infants**: open preferred. **Posterior lumbotomy** only in infants/toddlers; flank (11th–12th rib tip) any age.
- Laparoscopic = open (meta-analysis); robotic: shorter learning curve, faster resolution of hydronephrosis.

	Transperitoneal	Retroperitoneal
For	Easier landmarks; left transmesocolic	No bowel injury; leak extraperitoneal; shorter stay
Against	Bowel risk; urinary ascites → ileus	Learning curve; small space
Note	Preferred for horseshoe kidney, retrocaval ureter	One RCT favoured it; 80–90° flank, table **not flexed**, balloon 200–300 mL

- **Endopyelotomy**/balloon: poor long-term in < 18 years; niche = **failed pyeloplasty** (no vessel), double-J 6 weeks.
- Stones at pyeloplasty → flexible scope through a port.

5 · Stents, outcomes, failure

Stent	Removal	Point
Internal double-J	Second anaesthetic unless string/magnet	Catheter 24–48 h; string → more febrile UTI (girls)
Externalised nephro-ureteric	7–14 days, no anaesthesia	Fewer complications, shorter stay in infants

Stent	Removal	Point
Stentless	—	Needs flawless technique

- Stent duration 1–4 weeks does not change outcome. Antegrade double-J may not pass infant UVJ — check with methylene blue in bladder.
- Emergency diversion before repair: **nephrostomy, not retrograde stent** (inflames PUJ).
- Success 90–100% (variably defined). Commonest reason for conversion: bleeding.
- P1 Management of failed pyeloplasty — *early failure = oedema/clot → wait ± nephrostomy; late → endopyelotomy or redo transperitoneal pyeloplasty (best); ureterocalicostomy for short scarred ureter*

6 · Primary megaureter

- P3 Primary obst. megaureter. — *≥ 7 mm; four types; adynamic juxtavesical segment maturing craniocaudally; < 10 mm rarely needs surgery; indications incl. DRF < 40% or fall ≥ 5%; temporising options; excise–taper–5:1 reimplant; Starr/Kalicinski/Hendren; success > 90%*
- Hard part: separating obstructed from non-obstructed non-refluxing — dilatation does not prove obstruction. Follow into adulthood (some worsen at puberty).
- Infancy: small bladder vs massive ureter — safe non-refluxing tunnel impossible, so temporise. **Distal loop cutaneous ureterostomy** (inguinal; stomal stenosis rare); **refluxing reimplant** ("lesser evil"); balloon until waist disappears (weak, retrospective evidence; complications: migration, infection, VUR, stones); nephrostomy hard to maintain.

KEY POINT
Why not rely on the renogram? Neonatal washout depends on immature function, hydration and a huge ureter's capacity — serial DRF, diameter and infection history drive surgery. Megaureter is the PUJ problem moved to the bottom of the ureter: an immature adynamic segment that usually matures.

ALGORITHM
Excise stenotic distal segment → straighten → taper → non-refluxing reimplant at **5:1** tunnel length:diameter → stent (feeding tube 7 days or double-J 4 weeks)

- Plication/folding preserve blood supply; Hendren (excise lateral wall over 8–10 Fr catheter) risks ischaemic stricture — refinement excises only mucosa and muscle, sparing adventitia. Hendren: locking suture proximal two-thirds, interrupted distal third; taper gradually (abrupt calibre change obstructs).
- Very large ureter: extravesical approach avoids huge hiatus; laparoscopic/robotic comparable.
- POM does better than refluxing megaureter (stiff collagen-rich UVJ); worse with valves/neurogenic bladder.
- Complications: obstruction → stent first; ischaemic stricture → redo; reflux → observe/inject → revise; transureteroureterostomy for unilateral severe scarring.

7 · Surgery for vesicoureteral reflux

KEY POINT
Principles: exclude/treat bladder causes first · handle ureter (blood supply) and bladder (haematuria, spasm) gently · tension-free mobilisation · tunnel length:diameter **5:1** (Paquin) · straight course at empty and full bladder.

- Classified by access, approach (intra/extravesical) and tunnel vs original hiatus (supra/infrahiatal). Pfannenstiel; vertical midline for redo.

Technique	Approach / tunnel	Strength	Weakness
Politano–Leadbetter	Intravesical; suprahiatal neo-hiatus, vertical tunnel	Long tunnel for high grade	Blind neo-hiatus (bowel, vas, vagina); kink if too lateral/anterior
Glenn–Anderson	Intravesical; infrahiatal toward bladder neck	Least kinking	Short tunnel
Cohen	Intravesical; cross-trigonal	Commonest; success > **98%**; thick bladder; with bladder-neck reconstruction	Difficult later retrograde catheterisation
Lich–Gregoir	Extravesical detrusorotomy ≈ 5 cm	No haematuria/spasm; short catheter	Transient voiding dysfunction/retention up to **10%**, esp. bilateral
Robotic extravesical (RALUR-EV)	Y-shaped detrusorotomy 4–5 cm	Less pain, spasm, stay	Early series: leak, obstruction, ureteric injury

Cohen — key steps

ALGORITHM

Vertical cystotomy → feeding tube in orifice as handle → circumscribe orifice → free ureter (preserve distal blood supply) until it reaches contralateral wall tension-free → submucosal tunnel across trigone → new orifice above contralateral orifice → mucosa-to-mucosa 6-0, detrusor hitch at 6 o'clock

Later access to a cross-trigonal orifice: flexible ureteroscope, or suprapubic needle aimed at the orifice.

Lich–Gregoir — key steps

ALGORITHM

Bladder one-third full → divide obliterated umbilical artery → free ureter **hugging it**, only to the UVJ → mark 5-cm tunnel with bladder distended → detrusorotomy to mucosa → wide flaps → lay ureter in → close detrusor caudocranially, catching adventitia at apex → Foley 12–48 h

KEY POINT

Why voiding dysfunction? Pelvic plexus lies ≈ 1.5–2 cm dorsomedial to the UVJ. Stay on the ureter, avoid the dorsomedial UVJ, stop at the UVJ. Zaontz modification (detrusorotomy around UVJ) now avoided.

- AUA meta-analysis open success **95.9%**; minimally invasive converging.
- **Follow-up**: US at 6–12 weeks. VCUG may be omitted after low grade if US normal and no UTI/dysfunction; do it for high grade, new hydronephrosis, UTI or dysfunctional bladder.

Complications of reimplantation

	Early	Long-term
Reflux	Ipsilateral low grade usually resolves by 1 year. Contralateral ≈ 1 in 10 after unilateral repair, mostly resolves → observe; intervene for pyelonephritis. Routine bilateral repair not justified	Short tunnel/untapered ureter, or **unrecognised secondary reflux** (neurogenic/valve bladder) → treat bladder first (anticholinergic + CIC)
Obstruction	Oedema, clot, haematoma, kink within 2 weeks → double-J or nephrostomy; most resolve	Ischaemia, angulation (hiatus too lateral/anterior), tight tunnel, stenosis → balloon + stent → re-reimplant

- Early mild hydronephrosis should not exceed the maximal dilatation reflux caused.
- Contralateral reflux risk: age < 6 years, capacity < 50% expected, refluxing duplex; technique irrelevant.
- **Reoperation**: excise ischaemic ureter to bleeding peristaltic tissue; short ureter → **psoas hitch** (one side only), Boari flap, transureteroureterostomy, nipple valve, or Monti bowel ureter.

> **KEY POINT**
> Early complications are tissue reactions — drain and wait. Late ones are design errors (blood supply, hiatus position, tunnel ratio) or an untreated bladder; reimplanting into an abnormal bladder worsens dilatation.

Endoscopic injection

- P3 Endoscopic correction of VUR (STING procedure): indications and complications — *Dx/HA standard, ≈ 90% low grade; mound within 0.1–0.2 mL, volcano; never cauterise; STING vs HIT vs double HIT; late obstruction (Vantris); Swedish trial 20% recurrence at 2 years*
- Abandon if bladder inflamed. Failure → 1–2 repeats → reimplant (not harder after Deflux; implant encapsulated).
- Migration: particles < 50 µm reach systemic circulation; < 80 µm phagocytosed.

8 · Other ureteric lesions

- † **Retrocaval ureter**: right; behind IVC (persistent posterior cardinal vein); fish-hook/S → re-anastomose anterior to cava.
- **Congenital mid-ureteric stricture**: misread as megaureter; excise (1–3 cm) + ureteroureterostomy.
- **Fibroepithelial polyp**: upper third, male, left; pain + haematuria; retrograde or ureteroscopy; endoscopic polypectomy or resection.
- **Retroperitoneal fibrosis**: steroids → azathioprine → double-J → ureterolysis.

56 Hypospadias

1 · Definition, diagnosis and embryology

- **Triad**: proximal (ventral) ectopic meatus, ventral curvature, dorsally hooded foreskin (curvature and hood not always present). Incidence ≈ **1 in 150–300** male births; diagnosis clinical.
- **Megameatus with intact prepuce (MIP)**, ≈ 5%: wide distal meatus, normal foreskin — hidden until foreskin retracted; retract before every newborn circumcision.
- Avoid circumcision with curvature or ventral skin deficiency (foreskin may be needed).
- P3 Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) — *SRY → testosterone → 5α-reductase type 2 → DHT at 8–12 weeks; endodermal two-zipper vs outdated Glenister ectodermal theory; dorsal prepuce fusion first*
- Critical window **weeks 8–16**; closure proceeds from anus to glans tip. Two-zipper theory = Kurzrock (current). MCQ
- Ventral prepuce fusion awaits urethral fold fusion (≈ 20 weeks) → ectopic meatus has incomplete foreskin in **95%**.

> **KEY POINT**
> Hypospadias is an arrest of the androgen-dependent closing zipper: the meatus stops where closure stopped, the prepuce cannot close ventrally over an open urethra, and unfused dysplastic ventral tissue tethers the shaft into curvature.

2 · Aetiology

- **Multifactorial**; all converge on insufficient androgen action in weeks 8–16.
- **Fetal/placental**: prematurity, SGA, IUGR, placental insufficiency (≈ 2× risk); assisted reproduction (independent).
- **Genetic**: twin concordance 9–27%; first-degree relative 9–15%; chromosomal abnormality up to 30%; no single gene (HOX, FGF; WT1, SOX9, DMRT1, GATA4; LH receptor, 5α-reductase, androgen receptor). ≈ 200 syndromes: WAGR, Denys–Drash, Smith–Lemli–Opitz.
- **Endocrine disruptors** (pesticides, oestrogens/progestins, DES, phthalates, phytoestrogens): proven in animals, unproven in humans.
- **Testicular dysgenesis syndrome**: hypospadias, cryptorchidism, poor semen, testis cancer share one fetal exposure.
- **Anogenital distance (AGD)**: shorter = more severe hypospadias.

3 · Classification and severity

- Distal (glanular, coronal, distal shaft) **70–85%**; proximal 10–25%.
- Meatus is the **least reliable** marker — **classify after degloving**: meatus + curvature + plate.
- P4 Hypospadias: present concepts in management, and hypospadias fistula — *GMS score table (glans, meatus, shaft; total 3–12, higher → more complications); POST low ratio → inlay graft over plain TIP*
- GMS: Merriman 2013, scored in theatre.

4 · Associated anomalies and DSD work-up

- Hernia/hydrocele 9–16%; undescended testis (UDT) ≈ 7%, ≈ 10% proximal.
- **DSD work-up**: UDT + proximal hypospadias; perineal hypospadias; atypical genitalia. Since 2006 severe hypospadias = 46,XY DSD. DSD risk ≈ **50%** non-palpable vs 15% palpable testis; 64% proximal vs 7% distal.
- US for Müllerian remnants, karyotype, testosterone, FSH, LH, AMH. 46,XY does not exclude 5α-reductase deficiency, androgen insensitivity, testicular dysgenesis or ovotesticular DSD → *Ch 59*.
- **Prostatic utricle**: commoner proximal; large → stasis, UTI, difficult catheterisation; excision rare (vas at risk).
- No routine VCUG or renal US.

> **KEY POINT**
> Hypospadias + cryptorchidism signals a systemic androgen problem, not a local one — hence karyotype before surgery.

5 · Pre-operative assessment

- **Goals**: straight erection, urethra to tip, acceptable cosmesis, fewest procedures, durable. A straight penis voiding standing may gain little.
- Examine: stretched length (exclude micropenis), groins, AGD, penoscrotal transposition, torsion, scrotal cleft, meatal stenosis ("laser-beam" stream may justify surgery even when distal).
- **Timing**: genital awareness ≈ 18 months. AAP 1996: 6–12 months; authors **6–18 months**. Systematic review (>16,000): fewer complications when younger. One short anaesthetic low-risk.
- **Androgen**: glans width **<14 mm** → more complications. Testosterone cypionate **2 mg/kg IM at 5 and 2 weeks** pre-op if glans <15 mm (≈ 4 mm gain); DHT or hCG alternatives. Two RCTs fewer complications; meta-analyses inconclusive. Side effects (aggression, erections, pigmentation) resolve within 6 months.

6 · Intra-operative principles

- Caudal block (fistula link disputed), penile or pudendal block; ketorolac 0.5 mg/kg once; IV antibiotics. Tourniquet (release every 30 min) or epinephrine.
- Cystoscopy only for severe proximal/redo.
- Rapidly absorbed sutures (chromic, polyglactin) beat PDS; 6-0–8-0. Slow sutures → epithelialised tracks → fistula.
- Magnification, **tension-free** closure, non-overlapping suture lines, haemostasis. **Never hair-bearing skin** (stones, UTI). Higher surgeon volume → fewer complications.

7 · Curvature

Curvature, not meatal position, decides **one-stage vs staged**. Causes: ventral skin tethering, dysplastic dartos/Buck's fascia, short urethra, corporal disproportion.

ALGORITHM

Deglove to penoscrotal junction, excise dysplastic dartos → **artificial erection** (Gittes–McLaughlin, corporal compression or prostaglandin) → straight → urethroplasty · residual <30° → dorsal plication · >30° → divide urethral plate → still >30° → ventral corporal lengthening (usually staged)

- Mild <30°, moderate 30–45°, severe ≥45°. Eyeballing underestimates; tourniquet may hide basal curvature.
- Plication: Nesbit, Baskin midline 12 o'clock, Heineke–Mikulicz; ≤30° (authors ≤15°); risks nerve injury off midline, pubertal recurrence, shortening.
- Lengthening: transverse ungrafted corporotomies (STAG) or single 9-to-3 corporotomy + graft (dermis 20% larger, tunica vaginalis, SIS); risks graft contraction, limited ventral skin.
- Recurrence plication 28% vs grafting 9%; not dividing the plate is an independent risk. Correct fully first time.

8 · Urethroplasty

Correct curvature → urethroplasty → glansplasty → skin ± preputioplasty. TIP for >90% of distal repairs.

Technique	Indication	Key point / pitfall
MAGPI (Duckett 1981)	Glanular, mobile meatus, no curvature	Used more proximally → retrusive meatus
Urethromeatoplasty	Stenotic distal meatus	Heineke–Mikulicz
GAP	Wide deep groove (MIP)	Two-layer neourethra
Thiersch–Duplay	Wide plate	Tubularise over 12 Fr
TIP / Snodgrass (1994)	Distal, midshaft; narrow plate; mild curvature	Midline relaxing incision; stop 2–3 mm short; meatal stenosis; inlay graft if poor plate
Mathieu (flip-flap)	Distal, shallow plate	Narrow base → ischaemia, fistula; fewer stenoses than TIP
Onlay island flap	Plate kept but narrow	Diverticulum if too wide; less stenosis
Duckett tube (transverse preputial island)	Proximal; plate divided	Higher complications; diverticulum
Two-stage Bracka	Proximal severe curvature, redo, poor plate	Stage 1 quilted preputial/buccal graft; stage 2 (≥6 months) tubularise
Two-stage Byars	Proximal severe curvature	Dorsal prepuce flaps; highest staged complications

- **STAG**: lowest proximal complications (≈ 23%); STAC (3-stage, authors' choice); distal-first/Ulaanbaatar.
- Stage because urethroplasty over a fresh corporotomy/graft fails; repeat artificial erection at stage 2. Proximal TIP abandoned (devascularisation, recurrent curvature).

KEY POINT

Why proximal repairs fail — Poiseuille: resistance ∝ length / radius⁴. A long, rigid, spongiosum-free tube ending in a non-distensible glans raises proximal pressure → fistula, diverticulum, stricture. 25–75% need extra procedures after planned two-stage repair.

- Glansplasty: deep wings, tension-free **two-layer closure**. **Barrier layer** (Belman dorsal dartos, ventral dartos, **tunica vaginalis** — mobilise to external ring, spongiosum) reduces fistula; must not compress glans.

9 · Post-operative care

- Compressive dressing ≈ **72 h**. Diversion **3–21 days** by complexity (dripping stent; suprapubic for older/redo); short distal repairs unstented.
- Oxybutynin until stent out; stiff stents/lateral Foley pull → glans dehiscence. Paracetamol + ibuprofen; antibiotic prophylaxis evidence mixed.

10 · Complications

Complication	Standard definition
Fistula	Urethral leak below meatus
Glans dehiscence	Wing separation; glans fusion <2 mm
Meatal stenosis	Obstructive symptoms + **<8 Fr** prepubertal / <12 Fr postpubertal
Diverticulum	Segmental sacculation on voiding
Recurrent curvature	**>30°** on erection

- P4 Hypospadias: present concepts in management, and hypospadias fistula — *commonest complication (<10%); distal obstruction key; wait 6–12 months; calibrate, stress test, layered closure with barrier; coronal → redo urethroplasty; recurrence ≈ 20%*
- **Meatal stenosis**: dilatation ± steroid → meatotomy or redo; commoner after TIP than Mathieu.
- **Glans dehiscence**: reoperate only for voiding symptoms.
- **Stricture** (late): dilatation/urethrotomy rarely durable → buccal urethroplasty, often staged.
- **Diverticulum**: excise redundancy, keep dartos; exclude distal obstruction.
- **BXO/lichen sclerosus**: steroids usually fail → excise all, staged buccal/preputial graft; recurrence up to **40%**; can invade buccal grafts.
- **Recurrent curvature**: degloving + artificial erection; staged lengthening, Cecil if skin short. **Hairy urethra**: laser epilation or replace with buccal graft.

> **KEY POINT**
> Nearly every complication traces to distal obstruction, ischaemia or tension — so fistula/diverticulum repair begins by calibrating the distal urethra.

Reoperation

- Devine: avoid "small operations" — wide scar excision and tissue transfer. Always cystoscope and artificial erection.
- **Buccal mucosa** graft of choice; **lip mucosa** within glans; bladder mucosa abandoned (prolapse). Graft 10% larger, quilted; urethroplasty ≥6 months later. Dorsal inlay beats onlay/tube in redo.
- **Cecil–Culp**: bury penis in scrotum; release at 9–12 months. Nitroglycerin, hyperbaric O_2, expanders unproven.

11 · Outcomes and follow-up

- TIP meta-analysis (Pfistermuller 2015, 4,675 boys): complications 10.6%, reoperation 4.5%; fistula > stenosis > stricture; dartos helps. Distal success 83–95%; proximal complications 23–68%.
- ≈ 50% of complications in year 1, ≈ 75% by 3 years → follow proximal cases **through puberty**. Recurrent UTI (≈ 2%) → image for diverticulum, stricture, utricle.
- HOSE (surgeon), Penile Perception Score (patient/parent), HOPE (photos, cosmetic).
- Adults: untreated mild distal — few problems. Proximal — worse satisfaction, sensation, fertility, ejaculation; penile length chief complaint; no major psychopathology. Even 20–30° curvature can impair intercourse.

57 Etiology, Diagnosis and Management of the Undescended Testis

1 · Definitions

- P1 P3 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis; Maldescended (undescended) testes and treatment algorithm — *undescended vs ectopic (SIP commonest), retractile, acquired (8–11 years), vanishing (monorchia/anorchia) vs agenesis*

Term	Key point (beyond the papers)
Undescended testis (UDT)	Not in normal scrotal position (midpoint at or below mid-scrotum); palpable vs non-palpable (undescended or absent)
High scrotal testis	Stable, above scrotal midpoint — mixed group (stable, retractile, "gliding"); not routinely called undescended
Retractile	One that rarely stays down or rests high may progress to acquired UDT
Recurrent	Undescended at birth → descended spontaneously → re-ascended (10–22%, at 1–5 years)
Secondary / testicular retraction	Suprascrotal after hernia repair / after orchidopexy (iatrogenic)
Agenesis	Never formed; ipsilateral Müllerian persistence (extremely rare)

2 · Embryology of descent

- P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — *SRY/SF1 → SOX9 → Sertoli; INSL3 via RXFP2 + androgen; gubernaculum attaches to epididymis; transabdominal 10–15 weeks (outgrowth, CSL regression); scrotum by DHT at 10–12 weeks*
- Fetal Leydig cells active 6–7 weeks, involute after 24 weeks. INSL3 peak 15–17 weeks; testosterone peak 14–16 weeks (placental hCG, then fetal LH). AMH regresses Müllerian ducts 9–10 weeks. Sertoli cells lack androgen receptor early (else AMH suppressed, premature spermatogenesis). Humans have no cranial gonadal ligament (unlike rodents).

Phase	Timing	Event	Control
Transabdominal (relative)	10–15 weeks	Gubernacular swelling (outgrowth) from 12 weeks; cranial mesonephric ligament regresses by 13 weeks	INSL3 + androgen synergistic
Inguinoscrotal	**20–28 weeks**	Canal widens, processus forms; rapid transinguinal passage (rarely before 22 weeks); gubernaculum regresses; most scrotal after 27 weeks	Androgen via **GFN** → **CGRP**; cremaster myogenesis (Wnt)

> **KEY POINT**
> Outgrowth creates the canal; regression lets the testis follow — a defect in either gives the same phenotype. INSL3 and androgen act on one target (gubernacular/cremaster myogenesis), hence one clinical picture whichever fails.

Minipuberty (HPG surge, peak 1–3 months; earlier/stronger in preterm): LH, FSH, testosterone, AMH, inhibin B (INHB) rise; Sertoli cells proliferate (sets final size); gonocytes → **adult dark (Ad) spermatogonia** (stem-cell pool). Failed transformation → apoptosis or malignant change. Here a positional defect becomes a germ cell deficit.

3 · Epidemiology, aetiology and associations

- Prevalence **2–4% at birth** (term) → **1–2% at 3 months**; up to 45% preterm. Spontaneous descent mainly first 3–6 months, rarely after 6 months. High scrotal testes descend ≈ 88% vs extrascrotal ≈ half.
- 80–85% isolated. Risk factors: **prematurity/low birth weight**, family history (twins > brothers > sons; maternal half-brothers > paternal), maternal diabetes, smoking, DES (doubles risk), first-trimester corticosteroids in assisted reproduction, endocrine disruptors (mixture effects; unproven singly).
- Genetics: polygenic threshold trait; rare INSL3, RXFP2, NR5A1 variants.
- **Testicular dysgenesis syndrome**: cryptorchidism + hypospadias + infertility + TGCT. Short anogenital distance = low prenatal androgen; low placental weight associated.
- Non-syndromic UDT: primary gonadal dysfunction (raised LH/testosterone ratio, low AMH/INHB if bilateral) rather than central hypogonadism. No hormone predicts germ cell number → no routine testing outside DSD/anorchia.

Associated condition	Cryptorchidism	Implication
Prune-belly syndrome	All (bilateral abdominal)	Abdominal orchidopexy
Spigelian hernia	53–80%	Absent gubernaculum/canal
Cerebral palsy	25–54%	Often bilateral
Gastroschisis	5–25%	**50–55% descend spontaneously — observe**
Omphalocele, imperforate anus, myelomeningocele, PUV	12–33%	Caudal-field anomalies ipsilateral in 90%
Congenital diaphragmatic hernia	7–30%	Always ipsilateral
Klinefelter (47,XXY)	> 50% of prepubertal Klinefelter boys	< 2% of non-syndromic UDT; more with hypospadias
PMDS	Bilateral 59%; hernia uteri inguinalis; transverse ectopia 22%	AMH or AMHR2 mutation; adult infertility ≥ 80%, malignancy risk

Congenital vs acquired

Feature	Congenital	Acquired
Diagnosis	Birth/neonatal	Mean 8–11 years
Position	Often higher	Lower; often SIP
Processus vaginalis	Patent (87% unilateral)	Usually closed (patent 45–50%)
Epididymal anomalies	37–60%	11–31%

Acquired UDT = never-fully-descended testis unmasked by growth (reduced Ad spermatogonia, semen like congenital). Retractile testes share pathology; ascent ≈ 2–3% prospectively (higher in referral series) → **annual examination through childhood**. Screening (Japan) reduces late surgery.

4 · Histology, associated anomalies and fertility

- Germ cell loss begins **after infancy** and progresses; contralateral testis affected less. UDT smaller at birth and grows less even after spontaneous descent — intrinsic defect worsened

by position.

- Spermatogonia per tubule (S/T): 2.5 at birth, nadir 1.2 at 2–3 years (age-specific reference needed).
- Delayed gonocyte disappearance and Ad formation (Ad absent in 7–85%); primary spermatocytes (normally 4–5 years) delayed; Sertoli number falls after 4 months; reduced type IV collagen in both testes.
- Pathology similar whatever the position. Higher germ cell counts with younger age, palpable testis, less fibrosis.
- **Patent processus** 87% unilateral / 71% bilateral; aberrant lateral gubernacular attachment 66–75%; epididymal non-fusion > elongation/looping > atresia (track severity). UDT at higher torsion risk (mimics incarcerated hernia).

Anomaly	Key features	Management
Polyorchidism	Usually left, scrotal 75%; Kumar classification (A with vas, B without)	Observe normal scrotal testis; orchidopexy if undescended with intact drainage
Splenogonadal fusion	Left; continuous form with limb defects, micrognathia	Recognise — avoid needless orchidectomy
Transverse testicular ectopia	Hernia + contralateral non-palpable testis; PMDS 20–50%	Laparoscopy; trans-septal placement if vasa fused

Trial	Design	Result	Take-home
Kollin et al.	RCT: orchidopexy at 9 months vs 3 years	Early: more Sertoli and germ cells, better testicular growth (to 4 years)	Operate in first year; age at surgery, not hormones, predicts germ cells

Fertility outcome

Outcome	Former unilateral	Former bilateral
Oligospermia / azoospermia	31% / 14%	42% / 31%
Paternity	**≈ 90% of those attempting — as controls**	≈ 65% of those attempting (≈ 1/3 overall)

- Cryptorchidism = 10–17% of men evaluated for infertility.
- Unilateral: normal testis carries fertility; semen overstates infertility; paternity after orchidectomy of affected side = controls. Bilateral: semen reflects true infertility.
- Ad spermatogonia at biopsy predict adult counts (present both sides → normal count 84%; absent → all subnormal); Ad/tubule > 0.2 beats S/T. Bilateral: INHB/FSH and AMH/FSH ratios aid counselling.
- Even after first-year surgery 20–25% remain at infertility risk.
- Surgery **before 18 months** reduced assisted reproduction; each 6-month delay +5%.
- Orchidopexy before 19 years improves sperm retrieval in azoospermia; retrieval better than idiopathic NOA.
- Uncomplicated retractile testis: insufficient evidence of infertility.

5 · Malignancy

- TGCT risk **2–5×** (pooled OR ≈ 4); ≈ 1% of cryptorchid men; 5–10% of TGCT patients had UDT. Contralateral OR ≈ 1.5.

- **Swedish population study**: OR 3.2 — ≈ 2.8 if diagnosed before puberty vs ≈ 5.0 after. † Pettersson 2007: RR 2.23 if operated < 13 years vs 5.40 at ≥ 13. Meta-analysis: orchidopexy after 10–11 years → RR **5.8**.
- Untreated adults OR ≈ 14; risk rises with each 6-month delay beyond 18 months. Early surgery reduces the positional component only.
- Persistently cryptorchid → **seminoma** (74%); after orchidopexy → mostly non-seminoma (63%).
- GCNIS (from gonocytes failing to differentiate): 2–3% of formerly cryptorchid men, ≤ 0.4% of non-syndromic children; biopsy DSD/chromosomal cases. Contralateral GCNIS in 4–8% of men with TGCT (predictors: atrophy, cryptorchidism, age < 40).
- Biopsy at orchidopexy does not raise TGCT risk (earlier cohort confounded; Scandinavian universal-biopsy cohort: no excess).
- Postpubertal UDT, normal contralateral testis: **orchidectomy** (up to age 50) + prosthesis at same sitting.
- **Microlithiasis**: 3–6% of boys; not an independent TGCT risk → no routine US surveillance.

6 · Diagnosis

Examination — the gold standard

> **ALGORITHM**
> Warm room/hands, distraction; supine (then cross-legged, squatting, standing); thigh abduction inhibits cremaster; lubricant on fingers → sweep from ASIS along canal with one hand → palpate lowest position with the other → compress canal, invaginate scrotum to feel SIP → sustained traction fatigues cremaster: **stays = retractile; springs back = undescended**.

- Document palpability, position, size, mobility, lie, hernia/hydrocele, penile size, meatus. Scrotal asymmetry supports unilateral UDT.
- ≈ **80% palpable**; 60–70% unilateral. Non-palpable at surgery: abdominal/peeping 25–50%, vanished 15–40%, missed extra-abdominal 10–30%.
- Interobserver agreement poor; EUA is the final arbiter.
- P3 P1 Maldescended (undescended) testes and treatment algorithm; Vanishing testes — *AUA algorithm: bilateral non-palpable → DSD; re-examine at 6 months corrected age; EUA → yearly review / orchidopexy / exploration / laparoscopy; contralateral hypertrophy > 2 mL (specificity 100%) suggests monorchia, explore anyway*

Imaging — not recommended

AUA 2014: no US or MRI. US sensitivity ≈ 45%, specificity 78%; MRI 65%/100% (needs sedation). Neither avoids nor excludes surgery; only effect is ≈ 3 months' delay. Possible MRI use: ectopic abdominal testis not found at laparoscopy.

Bilateral non-palpable or UDT with hypospadias → DSD work-up

- Bilateral non-palpable newborn: **urgent karyotype and hormones** to exclude **CAH** (salt-wasting); pelvic US for Müllerian structures; defer circumcision.
- Hypospadias coexists in 12–24%; proximal hypospadias + UDT → chromosomal abnormality 32–47% *→ Ch 56*.
- Micropenis + bilateral UDT: hypogonadotrophic hypogonadism or anorchia (micropenis in nearly half) — LH, FSH, testosterone during minipuberty *→ Ch 55*.

Marker	Anorchia	Bilateral abdominal UDT
FSH, LH	Raised	Normal
AMH, INHB (best prepubertal Sertoli markers)	Undetectable/low	Present
Basal testosterone	Low in first year only	Normal
Testosterone after hCG	No rise	Rises

hCG stimulation poorly standardised, no longer test of choice; gonadotrophins may be normal in mid-childhood anorchia. **Laparoscopy remains the confirmatory test**. Genetic male with bilateral non-palpable testes: ≥ 95% abdominal.

7 · Management principles and timing

- **Timing**: observe to **6 months** (corrected) → refer → orchidopexy **6–18 months**, preferably by 12. Preterm: observe to 6 months beyond due date (1 year if marginal). Abdominal orchidopexy easier soon after minipuberty. Most children worldwide still operated after 2 years.
- Retractile: no surgery; yearly examination. Acquired UDT: operate (no paternity/cancer data to support observation, especially bilateral).
- Pubertal/postpubertal, normal contralateral: orchidectomy (laparoscopic if abdominal) ± prosthesis.

Hormonal therapy — not recommended (AUA)

- LHRH intranasal descent ≈ 20%; hCG never placebo-tested; neither separates retractile from undescended.
- hCG may increase germ cell apoptosis, reduce adult volume (infertility RR 4.7 after hCG in one series).
- Adjuvant LHRH (buserelin 10 µg intranasally alternate days × 6 months; Hadziselimovic) for bilateral UDT with absent Ad spermatogonia ("defective minipuberty"): pilot data only; needs biopsy.

> **KEY POINT**
> Hormones fail on both claims for one reason: position, not endocrinology, is the modifiable variable — descent is not androgen-limited after fetal life, and germ cells track age at scrotal placement.

8 · Surgery for the palpable testis

Inguinal orchidopexy — key steps

> **ALGORITHM**
> EUA → skin-crease incision → check SIP → open external oblique, spare ilioinguinal nerve → free testis, divide gubernaculum distal to sac (protect long-looping vas) → **separate patent processus from vas and vessels without skeletonising; high-ligate at internal ring** (main length gain) → divide lateral spermatic bands → retroperitoneal dissection through transversalis fascia → **Prentiss manoeuvre** (medial to inferior epigastric vessels) → transverse scrotal incision, **subdartos pouch** → pass testis untwisted → excise appendages, inspect epididymis → close dartos around cord (transalbugineal sutures unnecessary — 98.7% success without) → local/ilioinguinal/caudal block.

- Cannot reach scrotum → staged procedure; orchidectomy only for atrophic/abnormal testis, postpubertal, or insufficient vas.
- Biopsy not routine; indicated in DSD/dysgenesis; consider > 12–13 years and bilateral UDT.

Scrotal (Bianchi) orchidopexy

High scrotal-border incision → deliver testis → dissect sac/cremaster high, ligate processus via scrotal wound → subdartos pouch. Inguinal conversion 3.5–4.4%; atrophy 0.3–0.6%; less pain, similar efficacy — increasingly first choice for palpable (especially low) testes.

9 · The non-palpable testis

ALGORITHM

EUA — now palpable? → inguinal/scrotal orchidopexy. **Still non-palpable → diagnostic laparoscopy** (ring patency, vessels, vas, testis):

• **Blind-ending vessels** (± vas) above closed ring = vanished → stop (only unequivocal endpoint).

• **Vessels + vas entering ring** → inguinal/scrotal exploration; excise nubbin (viable tubules 5–15%).

• **Intra-abdominal testis** → primary laparoscopic or Fowler–Stephens (one/two stage); orchidectomy if postpubertal or dysgenetic.

• Peeping testis → laparoscopic or inguinal orchidopexy.

• Neither vas nor vessels → search perivesical area and retroperitoneum to the kidney (perirenal testis may accompany renal dysplasia/agenesis).

- Pitfalls: open ring usually (not always) means a viable distal testis; apparently blind-ending vessels may coexist with a distal/ectopic testis — if equivocal, keep exploring.
- Scrotal-first exploration with nubbin + contralateral hypertrophy: futile if vanished testis is intra-abdominal; "nubbin" may be a dissociated epididymis.
- Contralateral fixation of solitary testis: debated; prenatal torsion does not imply contralateral risk; not required by AUA.

Laparoscopic orchidopexy

ALGORITHM

Empty bladder/stomach → open (Hasson) umbilical 5-mm port, CO_2 8–12 mmHg → inspect both rings → 2–3-mm working ports → divide gubernaculum distal to ring (handle) → incise peritoneum lateral to vessels, distal to vas → mobilise vessels **preserving the peritoneal triangle between vas and vessels** (keeps FS possible) → new hiatus medial to inferior epigastric vessels, lateral to medial umbilical ligament → trans-scrotal port to subdartos pouch; avoid tension.

- Reaching opposite ring is unreliable test of length. Ring closure unnecessary. Contralateral patent processus ≈ 9% — repair optional.
- Also suits bilateral cases, abdominal wall defects, polyorchidism, splenogonadal fusion, transverse ectopia. Robot: feasible, no clear advantage.

Fowler–Stephens (FS) and alternatives

- P1 Methods of orchidopexy — *FS on vasal/gubernacular collaterals for testes not within 2–4 cm of ring; success table: primary laparoscopic > 90–97%, Jones/Bagley ≈ 95%, FS 60–97% (one-stage 63%, two-stage 68%), Shehata 84–85% (atrophy 10%), autotransplant 96%; Dessanti PTFE traction*

- Two-stage: clip vessels high (1.5–3 cm above testis) → wait ≈ **6 months** → mobilise on wide peritoneal pedicle. Two-stage slightly favoured (better position, similar atrophy).
- Keep peritoneum over vasal vessels; **spare the gubernaculum** (collaterals in 60%) — lowers atrophy.
- Avoid vessel division when possible (reduces germ cells, especially young boys). Retrovesical testis with short vas may need orchidectomy.
- Shehata: testis fixed near contralateral ASIS, orchidopexy after 12 weeks, keeps artery. Autotransplant for solitary abdominal testis (microsurgical skill).
- Bilateral high testes: stage sides ≈ 6 months apart.

> **KEY POINT**
> Every abdominal technique answers one question: how to gain length without losing blood supply. FS sacrifices the testicular artery; traction gains length over time while keeping it — vessel division is the last resort.

10 · Complications and follow-up

Complication	Cause / prevention	Management
Atrophy (≈ 2% modern; historically 24% abdominal/peeping vs 10% distal)	Vessel injury, skeletonisation, FS, tension	Offer **prosthesis** ≥ 6 months after scrotal surgery or after puberty, inguinal route, fixed to dartos
Retraction / non-scrotal (≈ 2–10%)	Inadequate retroperitoneal mobilisation, unligated processus	Redo: prescrotal → scrotal approach; else lateral approach through unscarred plane (Redman) or fascial patch on cord (Cartwright); results = primary
Vas injury	Long-looping vas cut with gubernaculum	Divide gubernaculum distal to sac; avoid skeletonising
Torsion after orchidopexy	Rare	Prevented by subdartos fixation
Laparoscopic	Bladder/vessel injury, hypercapnia, bowel obstruction	Hasson entry, bladder drainage

Follow-up: examine at ≥ **6 months** for position and size (US by some after FS); counsel on fertility and cancer; self-examination from puberty.

61 Urologic Considerations in Pediatric Renal Transplantation

- P1 Urological evaluation and management of pediatric kidney transplant patient. — *causes, five goals, pre-transplant screen (VCUG/urodynamics indications), bladder optimisation and cycling, augment segments, transplant technique, complication rates and management*

1 · Causes of ESKD in children

- **CAKUT** (hypoplasia/dysplasia), then obstructive uropathy — mainly **PUV** in younger boys; older: FSGS, neuropathic bladder. Commonest age at transplant **13–17 years**.
- Urologist joins the team **before ESKD**, not reactively.

> **KEY POINT**
> **Five goals**: unimpeded drainage · capacious low-pressure reservoir with social continence · volitional emptying · no infection · fewest operations. Chronic rejection is unavoidable, but every added insult accelerates it.

2 · Pre-transplant evaluation

- All: history + US + uroflow/PVR if toilet-trained. VCUG and urodynamics selective (neuropathic bladder, PUV, VUR, prior surgery, febrile UTI, worsening hydronephrosis); urodynamics repeated serially. Anuric child without urological history: nothing more.

> **ALGORITHM**
> Positive history → US + VCUG → abnormal dynamics → urodynamics → drugs + CIC ± bladder cycling → reassess → still unsafe → augmentation → clear for transplant

3 · Bladder management

- Commonest pattern: **low capacity, overactivity, poor compliance** — high pressure harms kidney even without reflux. Valve bladder evolves to **myogenic failure**.
- **Target**: expected capacity at detrusor pressure **< 40 cm H_2O**, socially continent.
- Antimuscarinics, mirabegron, botulinum toxin; α-blockers aid emptying; better storage may need CIC. CIC adherence predicts transplant-regimen adherence.
- **Defunctionalised bladder**: bladder cycling (rising saline volumes + drugs → urodynamics). **Never use the allograft to test a bladder.**
- **Augment** if cannot reach > **75%** expected capacity at < 40 cm H_2O with CIC no more often than every **3–4 h** despite maximal therapy; no clear rise in transplant complications. PUV → **postpone**; neuropathic → **earlier**.
- Segments: dilated ureter first (no mucus/stones/metabolic issues); ileum/sigmoid most used (malignancy worse with immunosuppression → lifelong surveillance); stomach rarely (haematuria–dysuria in anuric).
- **Mitrofanoff** sited **medial to planned transplant incision**; preserve native ureter for stoma, augment or graft drainage.
- Reconstruction **before** transplant, **6–12 weeks** healing; exception: transplant before 2 years (before continence expected).
- PD: intraperitoneal surgery forces temporary haemodialysis. Avoid concurrent procedures in infants. Long-term stoma/transplant ureterostomy only for severe neurodevelopmental

impairment or CIC intolerance.

- Circumcision in boys at risk (valves) reduces post-transplant UTI.

4 · Native nephrectomy

- P2 Pre transplant nephrectomy — *default keep; indications (hypertension, nephrotic, polyuria, infection/VUR, stones, malignancy, size); retroperitoneal on PD, embolisation if high risk; leave normal ureter, remove dilated refluxing ureter*
- Combined with transplant only for room (small child, huge kidneys) — ipsilateral via transplant incision.

5 · Transplant into the abnormal bladder

- Gibson retroperitoneal; < **15 kg** may need aorta/vena cava and transperitoneal approach. Know augment pedicle position.
- **Extravesical non-refluxing ureteroneocystostomy**: partially fill bladder → 3–4 cm detrusorotomy → spatulate 4–5 mm → interrupted mucosa-to-mucosa → close flaps loosely. Alternatives: Barry tunnel; **ureteroureterostomy** to native ureter (small bladder, short donor ureter, hostile pelvis; always the salvage).
- Stent if difficult: double-J out at **4–6 weeks**; early removal at 5–7 days may cut infection. Largest urethral catheter tolerated.

6 · Urological complications

Complication	Rate	Timing / management
Ureteric stenosis	**6%**	Most within 100 days; distal ischaemic; focal → balloon + stent 4–6 weeks; else open repair (native ureter, bladder flap) — do not delay
Urine leak	**3%**	Early; drain creatinine → catheter → US → MAG3/cystogram; small: observe; large: drain/explore; short necrosis → reimplant; long → native ureter, psoas hitch, flap
Significant VUR	2%	See below
Stones	2%	Treat if > 1 cm and growing, obstructing, infected; one-stage modality

MAG3 localises a leak only if graft function adequate. Pre-existing obstructive/reflux uropathy raises all complications.

Rising creatinine with hydronephrosis

> **ALGORITHM**
> Indwelling catheter 1–2 weeks → creatinine falls = bladder cause → if not, stent + catheter; rejection signs → biopsy; failed retrograde stent → nephrostomy + antegrade study

- Graft unusually sensitive to obstruction; hydronephrosis grade does not track impairment. Obstruction and rejection coexist — low threshold for diagnostic stenting.
- Extrinsic: lymphocele; **allograft mass = PTLD until proven otherwise**.
- Early UTI: catheter/stent; late UTI reflects bladder function. Colonisation universal with CIC — distinguish from infection.
- Graft VUR: febrile UTI → cystogram; recurrent febrile UTI + optimised bladder → **open/robotic reimplant** (intravesical, transtrigonal or Politano–Leadbetter; stent). Asymptomatic → observe. Endoscopic injection ≈ one-third success, obstruction risk.

7 · Outcomes

- Urological complications rarely lose the graft; thrombosis a leading cause of early loss.
- **≥ 2 UTIs** → lower graft GFR at 1 and 4 years, gap widening.
- PUV history → lower obstruction-free survival; graft-dysfunction link inconsistent.

63 Paediatric Urologic Oncology: Renal and Adrenal

1 · Wilms tumour (WT): epidemiology and genetics

- P2 Wilms tumour: classification, staging and management — *genes (WT1 11p13, IGF2 11p15), syndrome risks, screening, anaplasia, SIOP risk groups, COG staging, COG vs SIOP, nodes, adjuvant by stage*
- **6–7%** of childhood cancers, **95%** of kidney cancers under 15 y; median **3.5 y**; younger in bilateral/syndromic. Mostly somatic multi-hit (Knudson two-hit explains few); **10–15%** germline/early epigenetic; familial 1–2%.
- **Refer for genetics**: bilateral/multifocal, age < 2 y, rests, overgrowth/syndromic features, family history.
- WTX (Xq11) 20%, CTNNB1 15% — somatic only. TP53 4% (most anaplastic).
- WAGR = WT1 deletion, renal-failure risk into puberty; **Denys–Drash** = zinc-finger point mutation, **dominant-negative** (worse), early ESRD.
- **Type I** (WT1 ± CTNNB1): stromal, intralobar rests, early age, WAGR/DDS. **Type II** (11p15): blastemal/epithelial, perilobar rests, later, BWS.
- BWS: 21% bilateral; also hepatoblastoma, adrenocortical tumours; nephromegaly (≥ 95th centile) highest risk.
- 1p + 16q LOH (~5%) → **regimen M**; 1q gain (25–30%) next stratifier.

Syndrome risk	Examples
High (> 20%)	WAGR, DDS (**50%**); Fanconi D1 (BRCA2); Perlman (DIS3L2); mosaic variegated aneuploidy
Moderate (5–10%)	BWS, isolated hemihypertrophy, Frasier (WT1), Simpson–Golabi–Behmel (GPC3)
Low	Li–Fraumeni, NF1, Sotos, trisomy 18, Bloom

KEY POINT
Trap: BWS is the **commonest** predisposing syndrome but only moderate risk; WAGR/DDS are rarer and highest.

ALGORITHM
WT risk > 1% → renal US every **3–4 months** until **7 y** → suspicious lesion → CT/MRI

- BWS/hemihypertrophy/trisomy 18/SGB: US + AFP 3-monthly to 4 y (hepatoblastoma). Contralateral kidney after nephrectomy: US 3–4-monthly to 5 y (infants < 12 months with rests).
- Benefit = lower stage → NSS, not proven survival. Benign cysts common in BWS — avoid needless nephrectomy.

2 · Pathology

- Favourable: **triphasic**; soft, friable → spill risk; 12% multicentric unilateral.
- **Anaplasia**: nuclei ≥ **3×** neighbours, hyperchromasia, abnormal mitoses; ~2% under 2 y, ~13% over 5 y; acquired (absent from rests).
- Nephrogenic rests in > 1/3 of WT kidneys, **> 90%** of bilateral; most regress. Diffuse hyperplastic perilobar nephroblastomatosis: thick rind, almost always bilateral, ~half develop WT. Perilobar = periphery, late, BWS; intralobar = anywhere/sinus, early, WAGR/DDS.

3 · Clinical features and work-up

- > 85% symptomatic; gross haematuria → cystoscopy + retrograde at surgery (collecting-system extension). Persistent supine varicocele/hepatomegaly → IVC thrombus.
- **Acquired von Willebrand** up to 8% — clotting times may be normal; desmopressin.
- Chest CT before MRI anaesthesia (atelectasis). No contralateral exploration if imaging normal. Rupture: ascites beyond cul-de-sac best predictor; emergency surgery only if bleeding. Nodes cannot be judged on imaging. Imaging cannot separate WT from RCC/CCSK; CCSK/rhabdoid → bone scan + brain imaging.

4 · Staging and COG vs SIOP

- COG stage I–V (surgical–pathological): III = **positive nodes, any spill**, implants, positive margin, piecemeal removal. SIOP: pre-op VCR + AMD, no biopsy; follow-up US + chest radiograph (COG: CT).

> **KEY POINT**
> COG stage is surgical-pathological, so the surgeon — not the scan — decides chemotherapy intensity and irradiation: one spill converts stage I/II to III.

5 · Surgical principles

- Transperitoneal radical nephrectomy; inspect liver, peritoneum, nodes; intra-op US for equivocal vein/IVC.
- Node sampling (COG **≥5 nodes**, skip metastases interaortocaval); omission = commonest violation (10–20%) → under-staging.
- Spill ~10% (right side, large) → local relapse. Local recurrence risks: spill, unfavourable histology, incomplete removal, no node sampling.
- Complications ~11% (haemorrhage, SBO); more with > 10 cm, caval extension, thoracoabdominal incision, wrong diagnosis.
- Chemo: cyclophosphamide + etoposide in regimen M and CCSK; carboplatin for anaplastic/relapsed. † COG FH: I–II VCR + AMD; III–IV + DOX with flank/abdominal (and lung) RT.

Bilateral WT (stage V) and NSS

- P3 Bilateral Wilms tumour: treatment strategies — *5–7%; no upfront nephrectomy/biopsy; 3 drugs 6 weeks → NSS by week 12 (NWTS-4 anaplasia); open bilateral biopsy (discordant 4/5); enucleation + flank RT; transplant after 1–2 y; AREN0534 EFS 82%, OS 95%*

> **ALGORITHM**
> Synchronous bilateral → **no upfront nephrectomy, no biopsy** → VCR + AMD + DOX 6 weeks → imaging → NSS if feasible → else more chemo, **surgery by week 12** → non-responder: **bilateral open biopsy**

- Commonest cause of renal failure = bilateral nephrectomy for persistent tumour; remove residual renal tissue before transplant.
- Unilateral NSS: syndromic or solitary kidney; UMBRELLA **<300 mL**, remnant, negative nodes; frozen section, margin of normal kidney.

IVC thrombus

- ~6–10%; over half asymptomatic. Below hepatic veins → upfront nephrectomy, **en bloc** thrombectomy (piecemeal = stage III). **At/above hepatic veins → pre-op chemo** (avoid bypass); beyond 6–12 weeks adds nothing. Viable thrombus cells do not affect survival.

6 · Relapse and late effects

Relapse group	Definition	Salvage
Standard	Non-anaplastic, VCR/AMD only	70–80%
High	Non-anaplastic, after ≥ 3 drugs	40–50%
Very high	Anaplastic/blastemal	~10%

- Late effects track **RT and anthracycline**: heart failure 4.4% after DOX (rare below **150 mg/m²**); second cancers 1% at 10 y; gonadal failure. Renal failure after unilateral nephrectomy **0.6%** at 20 y (higher in WT1 syndromes).

7 · Other renal tumours

Tumour	Key facts	Management
Congenital mesoblastic nephroma	Commonest **neonatal** renal tumour (mean 3½ months); ETV6–NTRK3 (cellular type)	Radical nephrectomy alone; adjuvant only if incompletely resected
Clear cell sarcoma (CCSK)	3%; BCOR; **bone**/brain metastases; late relapse (10 y); never bilateral	DOX-containing chemo; RT even stage I
Rhabdoid (RTK)	2%; **most lethal**; median ~16 months; SMARCB1/INI1 loss (22q); brain tumours	No standard; tazemetostat on trial
Cystic nephroma / CPDN	Radiologically identical; CPDN has blastema in septa; DICER1	Excision cures; frozen section if partial
RCC	RCC > WT above 13–14 y; **translocation RCC (TFE3/TFEB, Xp11.2)** commonest; medullary carcinoma — sickle trait, lethal	Complete resection; nodes less ominous; chemo/radioresistant
Angiomyolipoma	Tuberous sclerosis (50–80%); bleeding risk > 3 cm and growing	mTOR inhibitor first; embolise for bleeding

> **KEY POINT**
> Age points to diagnosis: neonate → CMN; toddler → WT/rhabdoid; adolescent → RCC. Metastatic site sets staging tests: lung (WT), bone (CCSK), brain (rhabdoid/CCSK).

8 · Neuroblastoma

- Commonest **extracranial solid tumour** and commonest infant malignancy; neural crest; median 19 months; 50% adrenal; metastases ~70% at diagnosis. Familial < 1% (PHOX2B, ALK).
- Spectrum → ganglioneuroblastoma → **ganglioneuroma** (benign; operate only for symptoms). Shimada: stroma-poor vs rich, graded against age.
- Hard fixed mass, bone pain, periorbital ecchymosis, cord compression; pelvic → retention. Paraneoplastic: catecholamines, VIP → watery diarrhoea, **opsoclonus–myoclonus–ataxia** (half have neuroblastoma).

- Urinary **VMA/HVA** raised in **90–95%**. Marrow: two aspirates + two biopsies. MRI best for intraspinal extension. **Calcification and vessel encasement** favour neuroblastoma over WT.
- ^{123}I-MIBG required for staging (replaces bone scan); non-avid → PET; ^{131}I-MIBG therapeutic.
- Screening: more favourable infant tumours, no fall in advanced disease or mortality — **overdiagnosis**.

INRG stage	Definition
L1	Localised, **no** image-defined risk factor (IDRF)
L2	Localised, ≥ 1 IDRF (vessel/nerve encasement)
M	Distant metastases
MS	Age **<18 months**; skin, liver, marrow (< 10%) only; no bone (INSS 4S: < 12 months)

- INRG imaging-based, pre-treatment; INSS (1–4S) depends on resection/nodes — surgeon-dependent.
- Risk groups: stage, age, histology, MYCN, 11q, ploidy. **MYCN amplification** (> 10 copies, 2p24; ~20%) = high risk at any stage (except maturing histology); M at ≥ 18 months always high; MS with 11q aberration high.
- Adverse: 1p36 deletion, 11q LOH, 17q gain (strongest), **diploidy**. Favourable: **hyperdiploidy**, TrkA, age < 18 months, thoracic.

> **KEY POINT**
> Trap: diploid is bad, hyperdiploid good. Stage 4S/MS: OS 80–88%, frequent regression; deaths in infants < 2 months from massive hepatomegaly (respiratory compromise) or coagulopathy.

- **Low risk**: resection alone (stage 1 > 90% disease-free); chemo only for symptoms, progression, adverse biology. Never remove kidney/vital structures; RT to stage 2 residual unhelpful.
- **4S**: resection of primary not mandatory; asymptomatic favourable → supportive ± limited chemo.
- **Intermediate**: chemo then resection (fewer nephrectomies).
- **High risk**: induction (cyclophosphamide, vincristine, doxorubicin alternating cisplatin–etoposide) → delayed surgery after 3–4 courses → myeloablative consolidation + autologous stem-cell rescue (tandem better) → RT 15–30 Gy → **13-cis-retinoic acid + anti-GD2** (GM-CSF, IL-2).
- Stage 4: radical resection no survival benefit, better local control; delayed surgery = fewer ruptures/nephrectomies. Complications: diarrhoea (coeliac/SMA autonomic nerves), chylous ascites.
- **Cord compression**: chemo first; laminectomy only for progressive deficit (scoliosis); avoid RT.

> **KEY POINT**
> Surgical value runs inversely with biology: favourable tumours are cured by little or no resection; unfavourable ones are not rescued by radical resection — vital structures are never traded for margin.

9 · Adrenocortical tumours

- P4 Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — *bimodal (children < 10 y, 90% functional); Li–Fraumeni/BWS; open en bloc, intact capsule; mitotane + EDP*
- † Offer **TP53** testing/counselling to every child. Most functional, < 5 y: androgens → **virilisation** (commonest); cortisol → Cushing (often mixed); oestrogen → feminisation; aldosterone rare.
- † Work-up: DHEA-S, testosterone, androstenedione, 17-OHP, cortisol tests, **urinary steroid profile**, metanephrines; CT/MRI + chest CT.
- † Adverse: weight > 200 g, incomplete resection/spill, metastases, Cushing features; adenoma vs carcinoma hard to separate in children.
- † Open en bloc, **avoid capsular rupture**, no laparoscopy if carcinoma suspected; node dissection if large; **perioperative steroids** for cortisol-secreting tumours.

KEY POINT

Why open? Cure needs complete removal of an intact tumour; capsular rupture seeds the retroperitoneum, with poor salvage.

10 · Landmark trials

Trial	Design	Result	Take-home
AREN0532/0533	1p/16q LOH → regimen M	Stage III–IV EFS ~90% vs 61%	Biology-driven intensification
AREN0533 (lung)	Lung nodule response at week 6	RT omitted, EFS/OS preserved	Response-adapted RT omission
AREN0534	Bilateral WT: 3-drug, surgery by week 12	EFS 82%, OS 95% (vs 56%/81% NWTS-5)	Fixed clock, maximise NSS
COG P9641	Low-risk neuroblastoma, surgery alone	EFS ~89%, OS ~97%	Chemo only if symptomatic
COG anti-GD2	Anti-GD2 + cytokines + retinoid vs retinoid	Antibody superior	Immunotherapy standard in high risk
German NB97	Extent of resection, stage 4 > 18 months	No EFS/OS benefit	Radical resection not essential

64 Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

1 · Rhabdomyosarcoma (RMS): biology

- **Commonest childhood sarcoma** (~3% of paediatric cancers); 20–30% GU; peaks first 2 y and adolescence. Sites: bladder/prostate (BP), paratesticular (PT), vulvovaginal (VV). GU-RMS 5-y survival **>80%**. Metastases in 16% at diagnosis.
- ≥ 5% syndromic (usually embryonal): **Li–Fraumeni**, BWS, **DICER1**, NF1 (pelvic), Costello (bladder). Diagnosis < 3 y → seek germline predisposition.
- Surveillance: Li–Fraumeni → US 3–4-monthly + annual whole-body/brain MRI; DICER1 → chest CT + US; Costello → US 3-monthly to 8–10 y.

	Embryonal (ERMS)	Alveolar (ARMS)
Site	GU (commonest); **botryoid** from bladder/vagina	Trunk, extremities, perineum
Age	Younger (~70% < 10 y)	Older, adolescents
Fusion	Usually negative	~80% **PAX3/7–FOXO1** [t(2;13), t(1;13)]
Prognosis	Better	Fusion-positive worse; fusion-negative ARMS behaves like ERMS

KEY POINT
Fusion status predicts outcome better than histology, so trials assign risk by **fusion status**. MYOD1 mutation and circulating tumour DNA are emerging adverse markers.

2 · Presentation and work-up

- BP: haematuria, stranguria, retention, botryoid mass at meatus . PT: painless mass separate from testis. VV: bloody discharge, prolapsing mass.

ALGORITHM
US → CT/MRI of primary + nodal basin (BP: pelvic and retroperitoneal to renal vessels; PT: retroperitoneal; VV: inguinal, pelvic, retroperitoneal) → **chest CT** (lung commonest) → bone scan + bilateral marrow (omit in fusion-negative ERMS without nodal/lung disease)

3 · Stage, group and risk

Stage (pretreatment TNM, site) → **clinical group** (what surgeon left) → **risk group** (+ fusion, age).

COG stage	Site	Size / nodes
1	Favourable: **paratesticular, female GU**, orbit, head–neck	Any, M0
2	Unfavourable: **bladder/prostate**, extremity, parameningeal, trunk	≤ 5 cm, N0/NX
3	Unfavourable	≤ 5 cm N1, or > 5 cm any N
4	Any	M1

N1 = node ≥ 1 cm short axis. BP-RMS is never below stage 2.

Clinical group	Definition
I	Localised, completely resected, nodes negative
IIa / IIb / IIc	Microscopic residual / involved nodes resected, most distal negative / microscopic residual or most distal node positive
III	Gross residual or **biopsy only**
IV	Distant metastases

Risk	Criteria
Low	Fusion-negative; stage 1 group I–II (III orbit only), or stage 2 group I–II
Intermediate	Fusion-negative stage 1–2 group III (non-orbit), stage 3 group I–II; any fusion-positive non-metastatic; stage 4 fusion-negative < 10 y
High	Stage 4 fusion-positive, or ≥ 10 y

KEY POINT

Organ-preserving **biopsy** places BP-RMS in group III → **intermediate risk**; PT-RMS (resected by orchidectomy) is almost always low risk. Age < 1 y or > 10 y independently adverse. Trap: most distal node positive = IIc, not IIb.

4 · Treatment principles

- **Chemotherapy universal**; local control by RT and/or surgery; group sets RT dose. Complete resection (except PT) only if organ preserved.
- Primary exenteration fell from almost all (IRS-I) to **13%** (IRS-IV), OS unchanged (78–83%).
- **Pretreatment re-excision** (before chemo; resectable without organ loss, positive/uncertain margins): improves GU survival, can change group. **Delayed primary excision** (after chemo starts): no proven survival gain, cannot change group; only if R0 likely; may lower RT dose.

Bladder/prostate RMS

- Biopsy: **cystoscopic** preferred (resectoscope, minimal cautery) → percutaneous → open transvesical last (spill).
- Obstruction: **urethral catheter** first → JJ stent → nephrostomy. Avoid suprapubic tube (seeding); vesicostomy/ureterostomy delays chemo.
- **End-of-therapy residual mass**: usually stroma/mature rhabdomyoblasts → observe if stable; resection does not improve survival. Viable tumour after chemoradiation, M0 → partial cystectomy or cystoprostatectomy/exenteration can cure.
- Only ~40% "normal" bladders after preservation; **urodynamics at 6 months**, renal US annually ≥ 10 y.

Paratesticular RMS

- **Radical inguinal orchidectomy**. Prior scrotal approach → excise remaining cord inguinally; hemiscrotectomy/scrotal RT only for positive cord margin or skin invasion. Tumour abutting but not invading tunica vaginalis is **not** a positive margin.
- **Ipsilateral nerve-sparing RPLND** before chemo if **age ≥10 y** or radiological N1; ≥ 7–13 nodes to renal vessels.

KEY POINT
Why age, not CT? Older boys have far more retroperitoneal disease and CT has a high false-negative rate for nodes.

Female GU, perineal, infants

- Vaginoscopy + biopsy. Chemo ± brachytherapy controls most; no node dissection; hysterectomy/exenteration only if unresponsive. Cervical: loop biopsy, repeat after induction.
- Vulvovaginal (two-thirds of female GU): MRI + vaginoscopy after 3 and/or 6 cycles; positive residual → brachytherapy.
- Perineal/perianal: mostly alveolar fusion-positive; 5-y OS ~50%.
- Infants: RT often withheld → more local failure; RT deviation allowed in fusion-negative < 24 months.

Late effects: RT ~tenfold more sequelae; bladder dysfunction > 60%; sexual dysfunction, infertility, ovarian failure, vaginal stenosis; second cancers ~6×.

5 · Other bladder tumours

- **Urothelial**: mostly PUNLMP/low grade → TURBT, fulgurate base, US surveillance. **Recurrence → search for syndrome** (Lynch).
- **Spina bifida/exstrophy**: lifetime bladder cancer **2–4%**, aggressive, **not increased by augmentation**; screening asymptomatic patients ineffective — evaluate haematuria promptly.

Benign lesion	Key point	Treatment
Inflammatory myofibroblastic tumour	ALK rearrangement; intermediate malignancy	Complete resection
Nephrogenic adenoma	After trauma, infection, surgery, transplant	TURBT + treat cause; recurs
Eosinophilic cystitis	Biopsy essential; eosinophilia often absent	Steroids, antihistamines
Haemangioma	Severe haematuria; Klippel–Trenaunay, Sturge–Weber	Propranolol, laser; partial cystectomy

6 · Testicular tumours

- ~1/100,000 under 15 y; peaks 2–4 y and puberty; benign 38–74%. Prepubertal: **teratoma** commonest (benign); **yolk sac tumour (YST)** commonest malignant.
- Doppler US + LDH, **AFP**, β-hCG. US cannot reliably separate benign from malignant; cystic with echogenic foci → teratoma; **"onion-skin" → epidermoid cyst**.

	AFP	β-hCG
Source	**YST invariably**	Embryonal, mixed; rare prepubertal
Half-life	**5 days**	**24 h**
Caveat	High in infancy; adult level (< 10 ng/mL) by **6–9 months**	—

ALGORITHM
Inguinal approach → markers normal + benign-looking US → **testis-sparing partial orchidectomy** with frozen section → benign: close; germ cell malignancy or raised AFP → radical inguinal orchidectomy

- Prepubertal teratoma: partial orchidectomy, **no follow-up**. † Postpubertal teratoma malignant → radical.
- **YST**: radical inguinal orchidectomy; **90%** stage I; CT; AFP decline by half-life. Stage I → **surveillance, no RPLND/chemo**. Scrotal violation → stage II (chemo, no hemiscrotectomy). Nodes with normal markers → biopsy; nodes + raised markers = stage III; III–IV → cisplatin, etoposide, bleomycin. Persistent mass → biopsy/RPLND.
- **Stromal**: Leydig (commonest; precocious puberty; Reinke crystals ~40%), Sertoli, juvenile granulosa (benign in infants) → partial orchidectomy if markers normal; radical after Tanner 5. Chemo rarely needed.
- **Testicular adrenal rests** (poorly controlled CAH) → increase steroids first; partial orchidectomy if function threatened.
- Microlithiasis: no childhood surveillance.

DSD

Risk rises with Y material in dysgenetic gonads: complete gonadal dysgenesis ~35%, partial ~24%, Frasier, Denys–Drash, PAIS, Turner with Y = high; CAIS, mixed gonadal dysgenesis = intermediate; no Y = low. Gonadoblastoma (premalignant, often bilateral) → dysgerminoma. Gonadectomy individualised.

> **KEY POINT**
> Trap: **complete** androgen insensitivity is intermediate (not high) risk; partial is high.

7 · Other pelvic and genital tumours

- **Sacrococcygeal teratoma**: ~1/40,000, female; neurogenic bladder 36–51%, worse Altman II–IV. Annual renal–bladder US to toilet training; urodynamics for higher types. **Currarino triad** (anterior sacral mass, sacral defect, anorectal malformation) → neurogenic bladder.
- Paratesticular non-RMS (cystadenoma — VHL): testis-sparing, but inguinal + frozen section mandatory.
- Ovary: neonatal cysts regress; Sertoli–Leydig → **screen DICER1**.
- **Urethral prolapse** mimics RMS (doughnut swelling, prepubertal Black girls) → topical oestrogen/steroid.

★ Putting it together: common paediatric decisions

A · VUR after febrile UTI

ALGORITHM

1. US → VCUG if recurrent/admitted febrile UTI or abnormal US.

2. Treat BBD (constipation first).

3. Low risk (grade I–II, no BBD, boy, older) → observe ± CAP. High risk (grade ≥ III, girl, BBD, scars, infant) → CAP.

4. Breakthrough UTI, new scars, persistent high grade + poor function → injection or reimplant (→ *Ch 41, 53*).

B · Dilated upper tract

Condition	Key test	Operate when	Operation
PUJ obstruction	MAG3 diuretic renogram	Function falling or < 40%, worsening dilatation, symptoms, infection	Anderson–Hynes pyeloplasty
Primary megaureter	US, MAG3, VCUG	Falling function, infection, progression; most resolve	Tapered/folded reimplant; infants: diversion or balloon
Ureterocele / ectopic ureter	US, VCUG, functional scan	Obstruction, infection, incontinence (girls)	Incision, upper-pole heminephrectomy or UU (→ *Ch 52*)
MCDK	US + DMSA (non-functioning)	Rarely; most involute	Observe; check contralateral VUR/PUJ

C · Undescended testis

ALGORITHM

Palpable → orchidopexy at **6–18 months** (ideally by 12); no imaging.

Non-palpable → EUA → laparoscopy: blind-ending vessels → stop · vessels + vas entering ring → groin exploration · intra-abdominal → laparoscopic orchidopexy (primary/Fowler–Stephens).

Bilateral non-palpable or UDT + hypospadias → DSD work-up first → *Ch 59*.

D · Acute scrotum

	Testicular torsion	Appendage torsion	Epididymitis
Typical	Sudden pain, vomiting, high-riding testis	Prepubertal; upper-pole tenderness, "blue dot"	Gradual; urinary symptoms; seek anomaly
Action	**Explore at once** (Doppler misses partial torsion); fix both, remove if non-viable	Analgesia, rest	Culture; antibiotics if infected

E · Newborn with atypical genitalia

KEY POINT

Bilateral impalpable gonads + atypical genitalia = salt-wasting CAH until proven otherwise. Electrolytes, 17-OHP (after 36 h), karyotype; crisis days 5–7. Sex assignment/surgery not emergencies (MDT).

F · Wilms tumour vs neuroblastoma

	Wilms tumour	Neuroblastoma
Age	Median 3.5 years	† Younger (median ~18 months)
Imaging	Intrarenal († claw sign); may enter IVC	Calcification, vessel encasement; † crosses midline, displaces kidney
Markers / genetics	WT1, 11p15; WAGR, Denys–Drash, BWS	Urinary VMA/HVA raised 90–95%; MYCN amplification = high risk
Strategy	COG: nephrectomy first · SIOP: chemo first	By INRG risk: observe → surgery → multimodal